

Oxidized low-density lipoprotein potentiates angiotensin II-induced Gq activation through the AT1-LOX1 receptor complex: Implications for renal dysfunction

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This study provides **useful** in vitro evidence to support a mechanism whereby dyslipidemia could accelerate renal functional decline through the activation of the AT1R/LOX1 complex by oxLDL and AngII. As such, it improves the knowledge regarding the complex interplay between dyslipidemia and renal disease and provides a **solid** basis for the discovery of novel therapeutic strategies for patients with lipid disorders. The methods, data, and analyses partly support the presented findings, although the observed variability and need for further in vivo validation require additional research in this key area.

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Abstract

Chronic kidney disease (CKD) and atherosclerotic heart disease, frequently associated with dyslipidemia and hypertension, represent significant health concerns. We investigated the interplay among these conditions, focusing on the role of oxidized low-density lipoprotein (oxLDL) and angiotensin II (Ang II) in renal injury via G protein αq subunit (Gq) signaling. We hypothesized that oxLDL enhances Ang II-induced Gq signaling via the AT1 (Ang II type 1 receptor)-LOX1 (lectin-like oxLDL receptor) complex. Based on CHO and renal cell model experiments, oxLDL alone did not activate Gq signaling. However, when combined with Ang II, it significantly potentiated Gq-mediated inositol phosphate 1 production and calcium influx in cells expressing both LOX-1 and AT1 but not in AT1-expressing cells. This suggests a

critical synergistic interaction between oxLDL and Ang II in the AT1-LOX1 complex. Conformational studies using AT1 biosensors have indicated a unique receptor conformational change due to the oxLDL-Ang II combination. In vivo, wild-type mice fed a high-fat diet with Ang II infusion presented exacerbated renal dysfunction, whereas LOX-1 knockout mice did not, underscoring the pathophysiological relevance of the AT1-LOX1 interaction in renal damage. These findings highlight a novel mechanism of renal dysfunction in CKD driven by dyslipidemia and hypertension and suggest the therapeutic potential of AT1-LOX1 receptor complex in patients with these comorbidities.

Introduction

Dyslipidemia is a major risk factor of atherosclerotic heart disease in patients with chronic kidney disease (CKD). It has been postulated that the association between dyslipidemia and CKD is not solely a result of epidemiological comorbidities but rather a complex interplay of causality, where CKD exacerbates dyslipidemia, while dyslipidemia, in turn, contributes to the onset and progression of CKD.^{1,2} Since Moorhead et al. proposed the lipid nephrotoxicity hypothesis in 1982,³ accumulating evidence has suggested that increased plasma lipid levels contribute to the development of renal glomerular and tubular damage, primarily in animal models of dyslipidemia.⁴ The etiology of dyslipidemia-induced nephrotoxicity is complex and multifaceted, potentially involving the activation of certain cellular signaling pathways that culminate in renal injury through elevated levels of oxidized LDL (oxLDL).⁵⁻⁸ The lectin-like oxLDL receptor, LOX-1, is implicated in organ damage caused by dyslipidemia, and its expression is increased in hypertensive glomerulosclerosis.⁹ In murine models, a deficiency of LOX-1 leads to a reduction in renal dysfunction, which was precipitated by a systemic inflammatory state following significant myocardial ischemia and injury.¹⁰ This supports the hypothesis that LOX-1 plays a role in the development of inflammation-induced renal injury. However, to date, no studies have investigated the role of LOX-1 in nephrotoxicity caused by dyslipidemia.

Hypertension is an established risk factor for CKD, and it has been suggested that hypertension and dyslipidemia act synergistically to induce renal dysfunction.¹¹ The development of renal damage due to hypertension involves direct renal injury by vasoactive hormones, such as angiotensin II (Ang II), in addition to renal hemodynamic abnormalities associated with elevated body pressure.¹² We have shown that LOX-1 and the Ang II type 1 receptor (AT1) of the G protein-coupled receptor (GPCR) are coupled on the plasma membrane, and that G protein-dependent and β -arrestin-dependent AT1 activation mechanisms are involved in the signaling mechanism by oxLDL and the intracellular uptake of oxLDL, respectively.^{13,14}

Interestingly, it was recently demonstrated that AT1 exhibits different modes and degrees of G protein activation depending on the conformational changes that occur during activation by various ligands.¹⁵ Specifically, G protein α_q subunit (Gq)-biased agonists induce a more open conformation of AT1 than Ang II, resulting in a more potent activation of Gq, which is the primary mediator of Ang II-induced hypertension. In contrast, β -arrestin-biased agonists induce a closer conformational change, leading to the activation of β -arrestin without the activation of Gq.¹⁵ Indeed, we found that oxLDL selectively activates G protein α_i subunit (Gi) of AT1, without activating Gq,¹⁴ similar to that induced by β -arrestin-biased agonists.¹⁶ However, given that Ang II and oxLDL coexist in physiological environments, it is plausible to hypothesize that the effect of these ligands on AT1 in living organisms may differ from their effects when administered individually. Indeed, we found that the combination treatment of oxLDL and Ang II enhanced pro-inflammatory NF κ B activity compared to each treatment alone in cells overexpressing both AT1 and LOX-1.¹⁴ Based on our findings and the aforementioned structure-activity relationship of AT1, we hypothesized that the binding of both oxLDL and Ang II to LOX-1 and AT1, respectively, may result in a more open AT1 structure, leading to stronger downstream Gq signaling. Therefore,

this study aimed to investigate and clarify this hypothesis, focusing specifically on renal component cells, and ultimately demonstrate the *in vivo* relevance of this phenomenon in the development of renal injury or renal dysfunction under conditions of Ang II and oxLDL overload.

Results

Oxidized LDL potentiates Ang II-induced Gq signaling in a LOX-1-dependent manner

First, we examined the additive effect of oxLDL on Ang II-stimulated AT-1-Gq signaling in CHO (Chinese hamster ovary) cells that did not endogenously express LOX-1 and AT-1 but were genetically engineered to express these receptors (CHO-LOX-1-AT1) ¹³. **Fig. 1a** shows the dose-response curve of Ang II-induced IP1 production, which serves as a measure of Gq activity, in the presence of varying concentrations of oxLDL in CHO-LOX-1-AT1 cells. Consistent with previous findings, oxLDL alone did not stimulate IP1 production in CHO-LOX-1-AT1 cells ¹⁴. However, when oxLDL was supplemented at concentrations of 10 and 20 $\mu\text{g/mL}$ (but not at 5 $\mu\text{g/mL}$), it caused a similar leftward shift in the dose-response curve of Ang II, resulting in a more than 80% decrease in the Effective Concentration 50 (EC50) (EC50 values: 0 $\mu\text{g/mL}$, 9.40×10^{-9} M; 5 $\mu\text{g/mL}$, 5.21×10^{-9} M; 10 $\mu\text{g/mL}$, 1.68×10^{-9} M; 20 $\mu\text{g/mL}$, 1.55×10^{-9} M). The maximum IP1 production induced by Ang II in CHO-LOX-1-AT1 cells was unaffected by the addition of oxLDL.

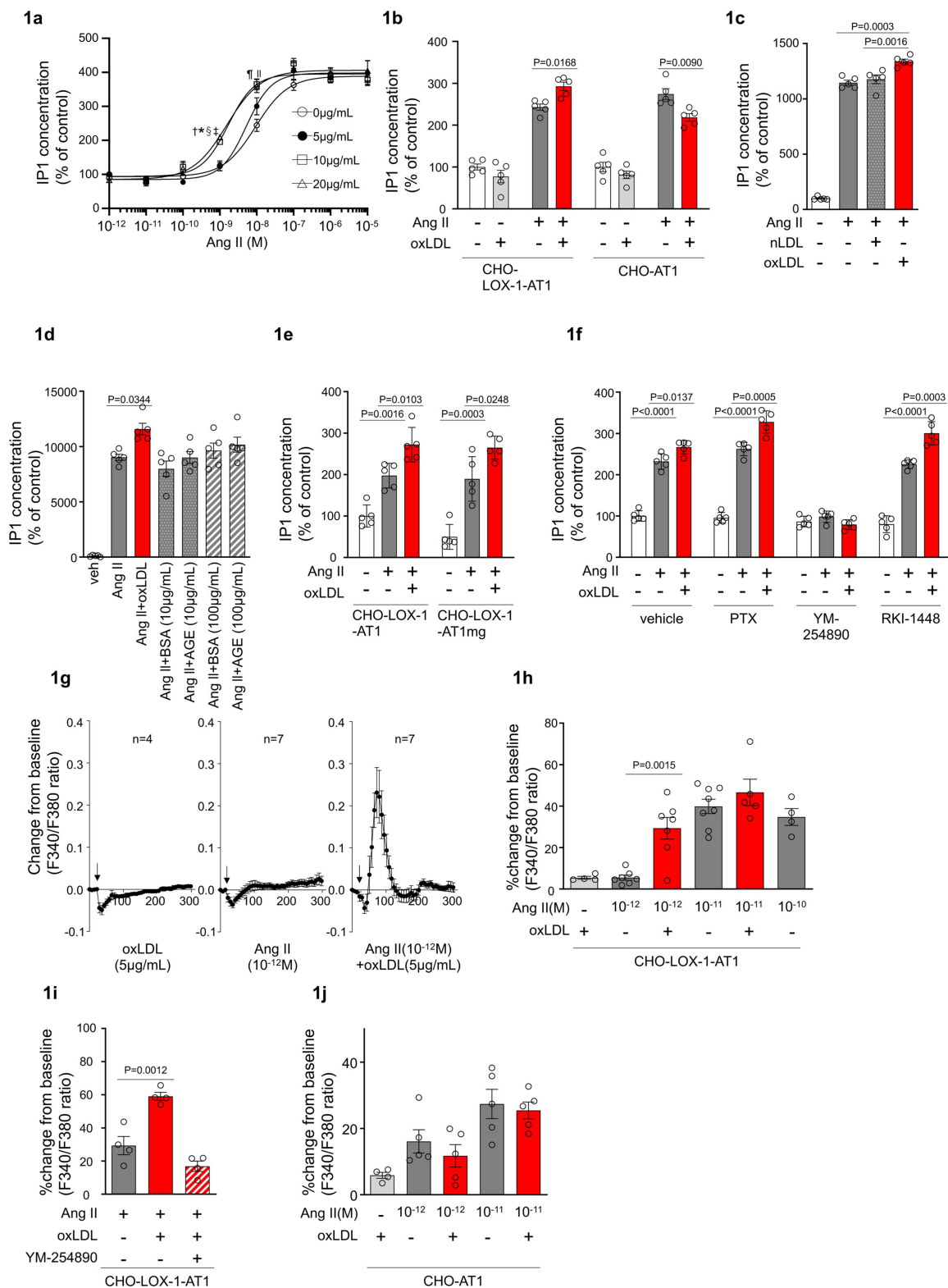


Fig. 1

| Oxidized LDL potentiates Ang II-induced Gq signaling and calcium influx in a LOX-1-dependent manner

a Dose-dependent response of IP1 concentration by the activation of Gq signaling in response to oxLDL and Ang II in CHO-LOX-1-AT1 cells. Cells were treated with oxLDL and Ang II at the concentrations described in the figure. (n=4 for each oxLDL concentration). * $P=0.0004$ for 0 $\mu\text{g/mL}$ vs 20 $\mu\text{g/mL}$, [†] $P=0.0003$ for 5 $\mu\text{g/mL}$ vs 20 $\mu\text{g/mL}$, [‡] $P=0.0020$ for 0 $\mu\text{g/mL}$ vs 10 $\mu\text{g/mL}$, and [§] $P=0.0015$ for 5 $\mu\text{g/mL}$ vs 10 $\mu\text{g/mL}$ at 10^{-8} M Ang II; ^{||} $P=0.0051$ for 0 $\mu\text{g/mL}$ vs 10 $\mu\text{g/mL}$, [¶] $P=0.0004$ for 0 $\mu\text{g/mL}$ vs 20 $\mu\text{g/mL}$ at 10^{-8} M Ang II.

b IP1 concentration in response to vehicle, native LDL (nLDL 10 $\mu\text{g/mL}$), and oxLDL (10 $\mu\text{g/mL}$) in the combination of Ang II (10^{-8} M) in CHO-LOX-1-AT1 cells (n=5 for each group)

c IP1 concentration in response to vehicle, oxLDL (10 $\mu\text{g/mL}$) in the combination of Ang II (10^{-8} M) in CHO-LOX-1-AT1 and CHO-AT1 cells (n=5 for each group).

d IP1 concentration in response to vehicle, oxLDL (10 $\mu\text{g/mL}$), BSA (10 or 100 $\mu\text{g/mL}$), BSA-conjugated AGE (10 or 100 $\mu\text{g/mL}$) in the combination of Ang II (10^{-8} M) in CHO-LOX-1-AT1 cells.

e IP1 concentration in response to vehicle, oxLDL (10 $\mu\text{g/mL}$) in the combination of Ang II (10^{-8} M) in genetically engineered CHO cells with or without intact β -arrestin binding domain (n=5 for each group). AT1mg indicates AT1 a mutant AT1 lacking a functional β -arrestin binding domain but retaining G-protein-biased signaling capability.

f IP1 concentration in response to oxLDL (10 $\mu\text{g/mL}$) in the combination of Ang II (10^{-8} M) and additional effect of PTX, a Gi inhibitor, YM-254890, a Gq inhibitor, and RKI-1448, a downstream Rho kinase inhibitor targeting G12/13 signaling, in CHO-LOX-1-AT1 cells (n=5 for each group).

g Intracellular calcium dynamics measured using Fura 2-AM by the ratio of the emission signals at excitation wavelength 340 nm and 380 nm in response to oxLDL (5 $\mu\text{g/mL}$), Ang II (10^{-12} M), and their combination (for each agonist, 4-7 regions of interest were selected). Addition of these agonists is marked with arrows on the timeline of the assay. **h** Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with oxLDL and Ang II at specified concentrations in CHO-LOX-1-AT1 cells as detailed in the figure (n=4-8).

i Percentage change from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM after stimulation with oxLDL (5 $\mu\text{g/mL}$), Ang II (10^{-12} M), and YM-254890, a Gq inhibitor, in CHO-LOX-1-AT1 cells.

j Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with oxLDL (5 $\mu\text{g/mL}$) and Ang II at specified concentrations in CHO-AT1 cells, as detailed in the figure.

Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a-f) and (h-j).

Additionally, we observed that oxLDL administration decreased Ang II-induced IP1 production in CHO cells expressing AT1 alone (CHO-AT1), in contrast to CHO cells expressing both AT1 and LOX-1 (**Fig. 1b**). This finding suggests that the potentiating effect of oxLDL on Ang II-AT1-Gq activity is dependent on the presence of LOX-1. However, the reason for the decrease in IP1 production caused by oxLDL supplementation was not determined in this study. Native LDL, which does not

bind to LOX-1, did not alter Ang II-induced Gq activity in CHO-LOX-1-AT1 cells (**Fig. 1c**). Furthermore, the presence of advanced glycation end products (AGEs) that bind to LOX-1¹⁷, did not enhance Ang II-induced IP1 production (**Fig. 1d**).

We found that enhanced IP1 production by co-treatment of oxLDL with Ang II was similarly observed in CHO-cells expressing LOX-1 and mutated AT1 with impaired ability to activate β arrestin¹⁴ (**Fig. 1e**). Moreover, the potentiating effect of oxLDL on IP1 production was unaffected by Pertussis Toxin (PTX), a Gi inhibitor, or RKI-1448, a downstream Rho kinase inhibitor targeting G12/13 signaling in CHO-LOX-1-AT1 cells. However, this phenomenon was completely inhibited by YM-254890, the Gq inhibitor (**Fig. 1f**). These results suggest that the potentiating effect of oxLDL on Ang II-induced IP1 production is not influenced by β -arrestin, Gi, or G12/13, which are the main effectors of AT1 signaling aside from Gq¹⁸.

oxLDL potentiates Ang II-induced calcium influx in a LOX-1-dependent manner

Calcium influx is a representative cellular phenomenon that occurs in response to Ang II-AT1-Gq activation. We found that oxLDL and low concentrations of Ang II (10^{-12} M) did not induce calcium influx in CHO-LOX-1-AT1 cells when treated alone, but apparently induced calcium influx when treated together (**Fig. 1g**, **h**). Ang II alone at higher concentrations (10^{-11} M) induced calcium influx, and no further enhancement was observed with oxLDL supplementation (**Fig. 1h**). The combined effect of oxLDL and Ang II on calcium influx was completely blocked by YM254890, suggesting that this phenomenon was Gq-dependent (**Fig. 1i**). Importantly, oxLDL supplementation with Ang II did not affect the calcium influx in CHO cells expressing AT1 alone (**Fig. 1j**).

Co-treatment of oxLDL with Ang II induces conformational change of AT1 different from each treatment alone

To gain mechanistic insight into this phenomenon, we initially conducted a live-imaging analysis of membrane LOX-1 and AT1 to determine whether the combined treatment of oxLDL and Ang II leads to increased internalization of AT1 upon activation compared to individual treatments, as described in a previous protocol paper¹⁹ (**Fig. S1**, Supplemental Video, and **Fig. 2a**).

Our findings revealed a decrease in green puncta, which represent AT1-eGFP, upon treatment with Ang II, oxLDL alone, and their co-treatment compared with the control group (**Fig. 2a**). However, the extent of membrane AT1 reduction was similar across all the treatment groups (**Fig. 2a**). Consistent with our previous report¹⁴, oxLDL treatment resulted in a reduction in the red puncta representing LOX-1-mScarlet (**Fig. 2a**). Importantly, co-treatment with oxLDL and Ang II did not further enhance the reduction of red puncta compared to oxLDL treatment alone (**Fig. 2a**). Based on these findings, it is conceivable that co-treatment with oxLDL and Ang II does not increase the content of activated AT1 compared to individual treatments alone.

AT1 intramolecular FAsH-BRET biosensors were used to detect AT1 conformational changes in CHO cells expressing LOX-1²⁰. Among several biosensors previously tested²⁰, we used two sensors with FAsH insertion at the third intracellular loop (ICL3P3) and cytoplasmic-terminal tail (C-tailP1) of AT1 that interacts with Renilla luciferase (RlucII) at the end of the cytoplasmic tail (AT1-ICL3P3 and AT1-C-tailP1, respectively), as these sensors were shown to enhance BRET signaling induced by Ang II compared to biased agonists, including SI, SII, DVG, and SBpA²⁰. In CHO cells expressing LOX-1 alone (CHO-LOX-1) transduced with lentivirus encoding AT1-C-tailP1, 10^{-5} M Ang II and the combination of Ang II and 10 μ g/mL oxLDL induced BRET similarly, whereas oxLDL alone did not alter BRET (**Fig. 2b**). In contrast, in CHO-LOX-1 cells transduced with AT1-ICL3P3-encoded lentivirus, the combination of Ang II and oxLDL induced BRET more prominently than Ang II alone, whereas oxLDL alone did not alter BRET (**Fig. 2c**). The difference

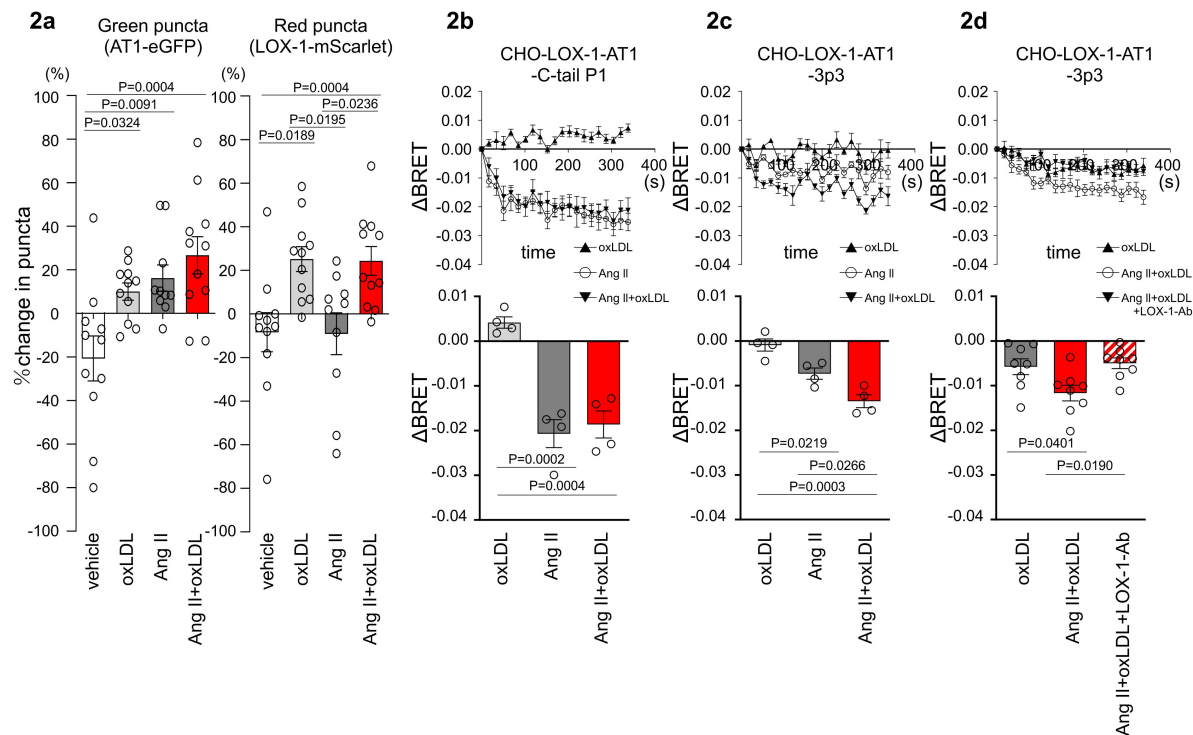


Fig. 2

| Co-treatment of oxLDL with Ang II induces conformational change of AT1 different from each treatment alone

a Change in green puncta (AT1-eGFP) and red puncta (LOX-1-mScarlet) by the treatment with vehicle, oxLDL (10 μ g/mL), Ang II (10⁻⁵M), and the combination of oxLDL and Ang II in CHO cell overexpressing these fluorescent protein-conjugated receptors ($n=10-11$ for each group). The puncta were manually counted by a blinded observer and the number of puncta at 0 and 3 min was determined (N0 and N3, respectively). The change in puncta was calculated as (N0-N3)/N0.

b-d Changes in BRET signals were monitored in CHO-LOX-1 cells expressing the following conformational biosensors: AT1-C-tailP1 (b) and AT1-ICL3P3 (c, d) bearing FIAsh insertion at the cytoplasmic-terminal tail (C-tailP1) and the third intracellular loop (ICL3P3) of AT1, respectively, that interact with Renilla luciferase at the end of the cytoplasmic tail. Cells were subjected to treatments with vehicle, oxLDL (10 μ g/mL), Ang II (10⁻⁵M), the combination of Ang II and oxLDL, and the combination of AngII, oxLDL and LOX-1 antibody. The BRET ratios were calculated every 16 seconds for a total of 320 seconds and the relative change in intramolecular BRET ratio (Δ BRET) was calculated by subtracting the average BRET ratio measured for cells stimulated with vehicle at each time point. Lower panels indicate average Δ BRET of all the time points during measurement. Data are represented as mean $\pm$ SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-d).

in sensitivity between the CHO-LOX-1-AT1-3p3 and CHO-LOX-1-AT1-C-tail P1 sensors likely explains why only the former showed a significant response to the combination of Ang II and oxLDL, underscoring the importance of FLAsH insertion site selection in these assays. Furthermore, the difference between oxLDL and the combination treatment was abolished by a neutralizing antibody against LOX-1 (**Fig. 2d**). These findings suggested that the concomitant binding of oxLDL to LOX-1 and Ang II to AT1 induced a conformational change in AT1 that was distinct from that induced by Ang II or oxLDL alone.

Oxidized LDL potentiates Ang II-induced Gq-calcium signaling in renal cells

To confirm the pathophysiological significance of this phenomenon observed in the overexpressing cells, we validated it in cells endogenously expressing LOX-1 and AT1. We found that oxLDL in combination with Ang II did not increase the cellular IP1 content in human umbilical vein endothelial cells (HUVECs), bovine vascular endothelial cells (BACEs), human aortic vascular smooth muscle cells (HAVSMCs), or rat macrophages (A10) (**Fig. S2**). In contrast, in normal rat kidney epithelial cells (NRK52E) and fibroblasts (NRK49F), the combination of oxLDL and Ang II, but not Ang II alone, increased IP1 accumulation, which was suppressed in the presence of YM-25480, the Gq inhibitor (**Fig. 3a, b**). IP1 accumulation induced by co-treatment with Ang II and oxLDL was abolished by the siRNA-mediated knockdown of either AT1 or LOX-1 (**Fig. 3c, d**; the knockdown efficacy is shown in **Fig. S3**). Regarding calcium influx, intracellular calcium concentrations were not increased by the treatment of either 10^{-7} M Ang II or low concentration of oxLDL (2 μ g/ml) alone (**Fig. 3e**, left and middle, **Fig. 3f**). In contrast, the combination of Ang II (10^{-7} M) and oxLDL (2 μ g/ml) increased intracellular Ca^{2+} concentration (**Fig. 3e**, right, **Fig. 3f**). The combined effect on calcium influx was attenuated by the siRNA knockdown of either AT1 or LOX-1 (**Fig. 3g**). The Gq inhibitor and angiotensin receptor blocker (ARB), Irbesartan, also inhibited this phenomenon (**Fig. 3h**). Calcium influx was not induced by either combination therapy or monotherapy in NRK52E cells (**Fig. S4**).

Co-treatment of oxLDL and Ang II enhanced cellular response upon Gq activation in renal cells

In NRK49F cells, co-treatment of oxLDL and Ang II, compared to vehicle, increased mRNA levels of NADPH components, *p67phox* and *p91phox*, fibrosis markers, *fibronectin*, *collagen-1*, *collagen-4*, and *TGF β* , and inflammatory cytokines, *TNF α* , *IL1 β* , *IL-6*, and *MCP-1* (**Fig. 4a**). Notably, oxLDL did not alter the expression of the genes of interest, and Ang II increased the mRNA levels of only *fibronectin* and *MCP-1*, indicating the apparent synergistic effect of co-treatment with Ang II and oxLDL on specific gene expression. This amplification effect of co-treatment was also observed in limited genes including *p67phox*, *TGF β* , *IL-6*, and *MCP-1* in NRK52E cells (**Fig. 4b**). The combined effects of oxLDL and Ang II on gene expression were completely abolished by the Gq inhibitor (**Fig. 4c, d**) and ARB (**Fig. 4e, f**) in NRK49F and NRK 52E. We also verified protein expression of α SMA as a molecular marker of epithelial-mesenchymal transition (EMT) upon the indicated stimulation for 3 days in rat kidney cells. The combination of Ang II (10^{-7} M) and oxLDL (5 μ g/ml) induced α SMA expression in NRK49F or NRK52E to the same extent or less than TGF β , a major inducer of EMT, respectively (**Fig. 5a, b**). The results of the combination treatment were strikingly different from that of oxLDL or Ang II treatment alone, which did not affect α SMA expression in both types of cells (**Fig. 5c, d**). The induction of α SMA by the combined treatment was suppressed in the presence of a Gq inhibitor or ARB in both types of cells, suggesting the AT1-Gq-dependent pathway in this phenomenon (**Fig. 5e, f** and **Fig. 5g, h**). As a final in vitro assay, the proliferative activity of NRK49F cells after 24 h of treatment with Ang II and oxLDL, either alone or in combination, was measured using the BrdU assay (**Fig. 6a**). When administered alone, Ang II and oxLDL increased and reduced BrdU incorporation, respectively. The reducing effect of oxLDL on proliferation was blocked by Irbesartan, but not by a

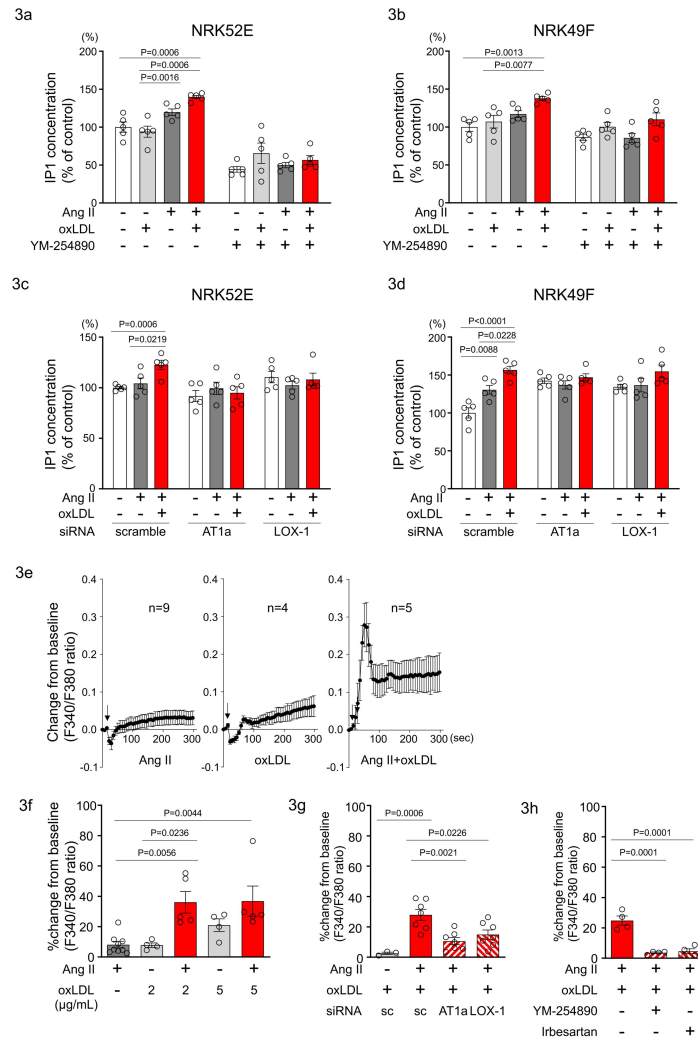


Fig. 3

| Oxidized LDL potentiates Ang II-induced Gq-calcium signaling in renal cells

a, b IP1 concentration in response to Ang II (10^{-7} M), oxLDL (10 μ g/mL), and the combination of both with or without YM-254890, Gq inhibitor, in NRK52E (a) and NRK49F cells (b) (n=5 for each group).

c, d IP1 concentration in response to Ang II (10^{-7} M) and oxLDL (10 μ g/mL) in the combination of Ang II (10^{-7} M) and the additional effect of siRNA-mediated knockdown of *AT1a* or *LOX-1* in NRK52E (c) and NRK49F cells (d) (n=5 for each group).

e Intracellular calcium concentration in NRK49F cells using Fura 2-AM and dual-excitation microfluorimetry. Changes in the fluorescence intensity ratio (F340/F380) served as an index of the calcium dynamics. Cells were exposed to Ang II (10^{-7} M), oxLDL (2 μ g/mL), and a combination of both agents. Addition of these agonists is marked with arrows on the timeline of the assay. Data acquisition and analysis were performed using a digital image analyzer to monitor real-time calcium flux (n=4-9).

f Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL at the concentrations detailed in the figure in NRK49F cells (n=5 for each group).

g Impact of siRNA-mediated knockdown *AT1* or *LOX-1* on the percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL (2 μ g/mL) in NRK49F cells (n=3-7).

g Impact of co-treatment of YM-254890, Gq inhibitor, or Irbesartan, an angiotensin receptor blocker (ARB), on the the percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL (2 μ g/mL) in NRK49F cells (n=4 for each group).

Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a-d) and (f-h).

Gq inhibitor, consistent with our findings that oxLDL induces biased activation of AT1, which favors Gi but not Gq signaling¹⁴. Oxidized LDL potentiated the Ang II-induced increase in BrdU incorporation, which was blocked in the presence of a Gq inhibitor or an ARB (Fig. 6a). The siRNA-mediated knockdown of AT1 completely abolished the effects of Ang II and oxLDL, either alone or in combination. Knockdown of LOX-1 did not affect the pro-proliferative effect of Ang II itself but blocked the enhanced proliferation induced by co-treatment with oxLDL (Fig. 6b).

Oxidized LDL-inducible diet exacerbates Ang II-induced renal dysfunction in wildtype mice, but not in LOX-1 knockout mice

To examine the relevance of this phenomenon in renal injury, we replicated the *in vivo* conditions that occurred during simultaneous stimulation with oxLDL and Ang II in wildtype (WT) and LOX-1 knockout (LOX-1 KO) mice. This was achieved by inducing oxLDL via a high-fat diet (HFD)²¹ and Ang II via a subcutaneously implanted osmotic mini-pump (Fig. 7a). Two different doses of Ang II were used in the experiment: a pressor dose of 0.7 μ , which was demonstrated to increase blood pressure (BP) and induce renal dysfunction^{22,23}, and a subpressor dose of 0.1 μ , which does not affect BP. For the intended analysis, the results are presented separately for mice exposed to subpressor and pressor doses of Ang II, utilizing the same outcomes of control animals infused with saline for comparison. Consistent with a previous report²⁴, the HFD used in the study prominently increased the plasma LOX-1 ligand concentration, which was undetectable under a normal diet (ND) after 6 weeks of feeding (Fig. S5). Notably, the HFD did not intensify but rather attenuated the body weight increase during the experimental period compared to the ND (Fig. S6a-c), likely due to reduced food intake in mice fed the HFD used in this study (Fig. S6d, e). The pressor dose of Ang II similarly increased BP in HFD-fed and ND-fed WT mice (Fig. 7b, Fig. S6f). However, notably, there was a modest trend of attenuated BP elevation by 0.7 μ Ang II infusion in LOX-1 KO mice, in line with the previous report showing reduced Ang II-induced BP elevation by LOX-1 deficiency in mice (Fig. 7b, Fig. S6f)²⁵. As expected, a subpressor dose of Ang II did not alter BP in the corresponding mouse group (Fig. 7b, Fig. S6g). Regarding biofluid analysis, HFD with saline infusion did not alter the urinary 8-OHdG concentration as a marker of oxidative stress and urinary albumin excretion (UAE) compared to ND with saline infusion in either WT or LOX-1 KO mice (Fig. 7c, d). The pressor dose of Ang II with ND significantly increased the urinary 8-OHdG concentration and UAE in WT mice (Fig. 7c, d). Notably, HFD feeding in WT mice with a pressor dose of Ang II resulted in a prominent increase in urinary 8-OHdG concentration and UAE compared to mice fed with ND (Fig. 7c, d). In WT mice administered with a subpressor dose of Ang II, a significant increase in UAE was observed when comparing the effects of HFD to ND (Fig. 7d). There were no significant differences in urinary 8-OHdG levels between the two dietary conditions (Fig. 7c). Interestingly, when LOX-1 KO mice were fed either HFD or ND and then administered the corresponding dose of Ang II, no differences were observed in the measured parameters (Fig. 7c, d). These findings indicate that the combination of HFD and Ang II administration appears to have a more pronounced effect on certain biofluid markers of renal injury in WT mice than in LOX-1 KO mice. The presence or absence of LOX-1 appears to influence the interaction between HFD and Ang II, affecting these specific parameters in mice. We did not find a difference in plasma aldosterone concentration between ND- and HFD-fed WT mice with a pressor dose of Ang II (Fig. S7).

We then conducted quantitative real-time PCR analysis on genes within kidney sections, encompassing NADPH components (*p40phox*, *p47phox*, *p67phox*, and *p91phox*), inflammatory cytokines (*IL-6*, *TNF α* , *IL1 β* , *MCP-1*, and *COX-2*), fibrosis markers (*TGF β* , *fibronectin*, *collagen-1 α* , *collagen-4 α* , *α SMA*, and *vimentin*), epithelial markers (*E-cadherin* and *cadherin-16*), and tubular marker (*NAGL*) (Fig. 8, Fig S8). In mice treated with a pressor dose of Ang II, 15 out of 18 genes examined showed enhanced alterations in gene expression, indicative of heightened NADPH oxidase components, inflammatory cytokines, fibrosis, decreased epithelial markers, and exacerbated tubular injury in HFD-fed WT mice compared to ND-fed mice (Fig. 8a, Fig. S8a).

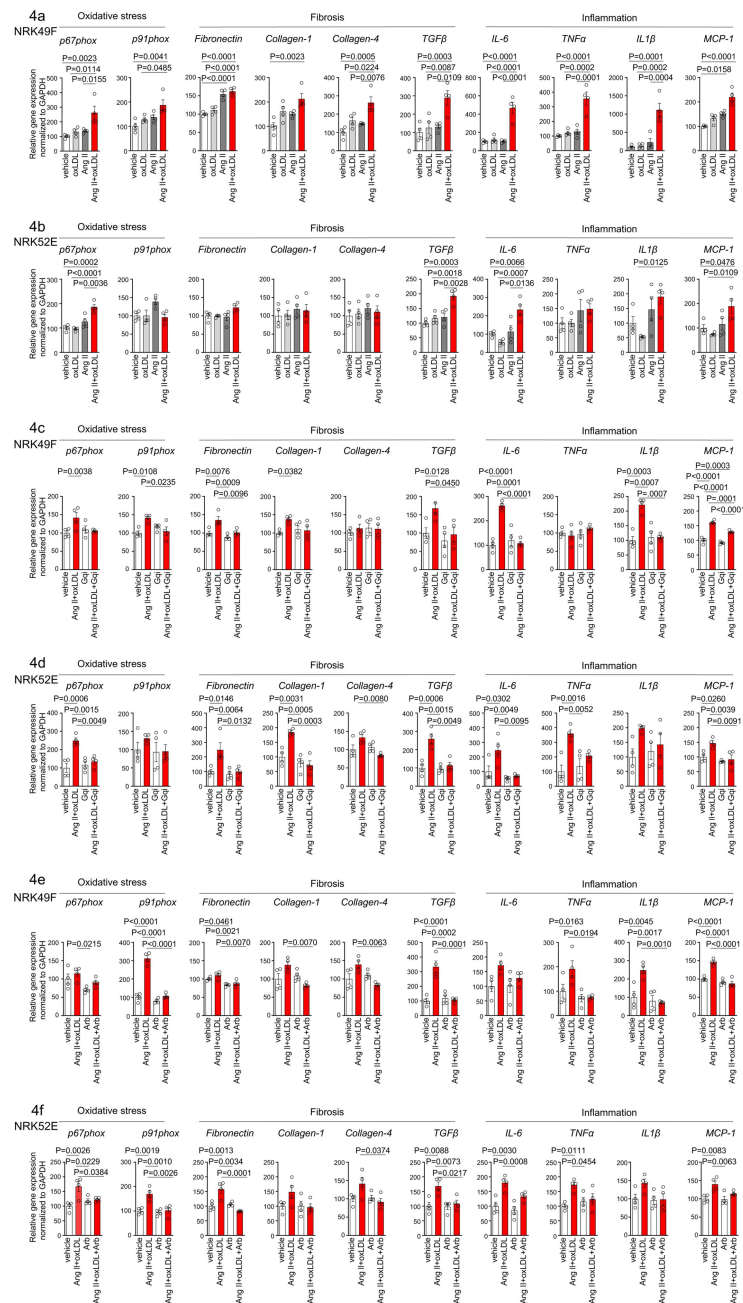


Fig. 4

Co-treatment of oxLDL and AII enhanced cellular response upon Gq activation in renal cells

a,b Quantitative real-time PCR analysis was performed to measure the gene expression of NADPH oxidase subunits (*p67phox* and *p91phox*), fibrosis markers (*fibronectin*, *collagen-1*, *collagen-4*, and *TGFβ*), and inflammatory cytokines (*TNFα*, *IL1β*, *IL-6*, and *MCP-1*) in NRK49F cells (a) and NRK 52E cells (b). Gene expression levels were normalized to those of GAPDH. Cells were stimulated by oxLDL (5 μg/mL), Ang II (10⁻⁷ M) or their combination (n=4 for each group).

c, d Cells were pre-treated with vehicle or Gq inhibitor (YM-254890, Gqi), followed by treatment with vehicle or the combination of oxLDL (5 μg/mL) and Ang II (10⁻⁷ M) in NRK49F cells (c) and NRK 52E cells (d) (n=4 for each group).

e, f Cells were pre-treated with vehicle or ARB (Irbesartan, Arb), followed by treatment with vehicle or the combination of oxLDL (5 μg/mL) and Ang II (10⁻⁷ M) in NRK49F cells (e) and NRK 52E cells (f) (n=4 for each group).

Data are represented as mean ± SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-f).

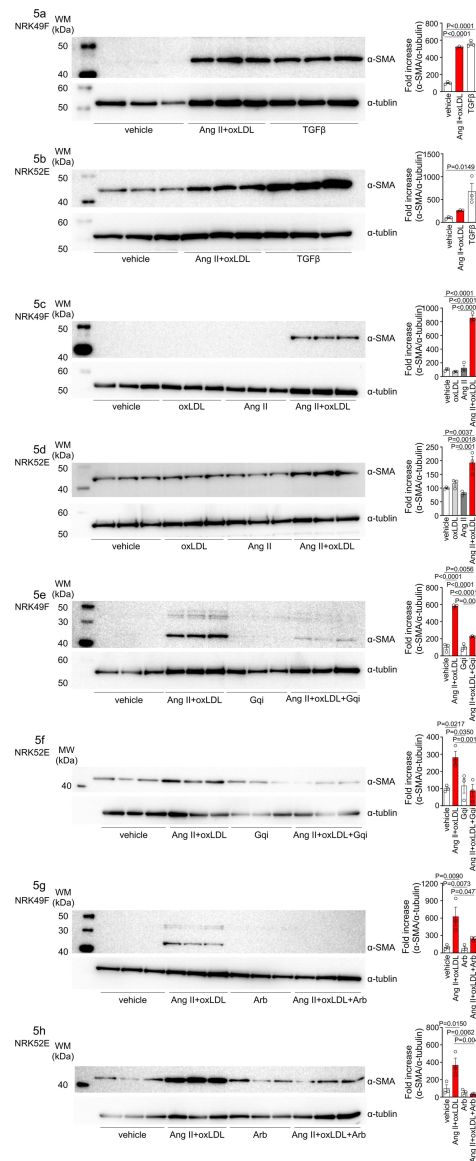


Fig. 5

| Oxidized LDL enhanced Ang II-induced epithelial mesenchymal transition in NRK52E and NRK49F cells

a, b Left: Western blot analysis of α-smooth muscle actin (α-SMA), a marker of epithelial-mesenchymal transition (EMT), in NRK49F (a) and NRK52E (b) cells. Cells were stimulated with oxLDL (5 μg/mL), Ang II (10⁻⁷ M), and TGF-β (10 ng/mL), with TGF-β serving as a well-known EMT inducer. Right: Densitometric analysis of α-SMA protein expression normalized to α-Tubulin (n=3 for each group).

c, d Left: Western blot analysis of α-SMA in NRK49F (c) or NRK52E (d) after stimulation with oxLDL (5 μg/mL), Ang II (10⁻⁷ M), and their combination. Right: Densitometric analysis of α-SMA protein expression normalized to α-tubulin (n=3 for each group).

e, f Left: Western blot analysis of α-SMA in NRK49F (e) or NRK52E (f) after treatment with a combination of oxLDL (5 μg/mL) and Ang II (10⁻⁷ M). Prior to this treatment, the cells were pre-treated with either a vehicle or a Gq inhibitor (YM-254890, Gqi). Right: Densitometric analysis of α-SMA protein expression normalized to α-Tubulin (n=3 for each group).

g, h Left: Western blot analysis of α-SMA in NRK49F (g) or NRK52E (h) after treatment with a combination of oxLDL (5 μg/mL) and Ang II (10⁻⁷ M). Prior to treatment, cells were pre-treated with either vehicle or ARB (Irbesartan, Arb). Right: Densitometric analysis of α-SMA protein expression normalized to α-tubulin (n=3 for each group).

Data are represented as mean ± SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-f).

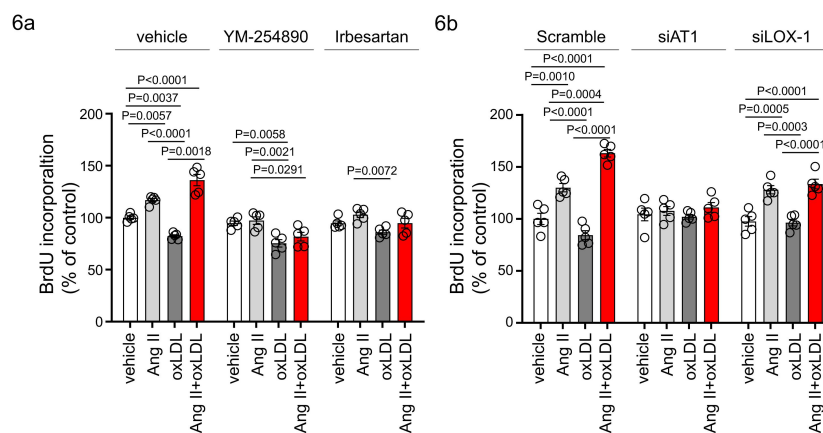


Fig. 6

| Oxidized LDL enhanced Ang II-induced renal fibroblast proliferation via AT1-Gq signaling and LOX-1-dependent manner

a Proliferative activity assessed by BrdU incorporation into NRK49F cells. Cells were pretreated with vehicle, YM-254890, or ARB, Irbesartan, followed by the treatment with oxLDL (5 µg/mL), Ang II (10^{-7} M), or their combination. (n=5 for each group).

b NRK49F cells were subjected to siRNA-mediated knockdown using specific siRNAs for *AT1a* (siAT1) or *LOX-1* (siLOX-1). Following knockdown, cells were treated with either vehicle, oxLDL (5 µg/mL), Ang II (10^{-7} M), or their combination. Proliferative activity was assessed by measuring the BrdU levels (n=5 for each group).

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).

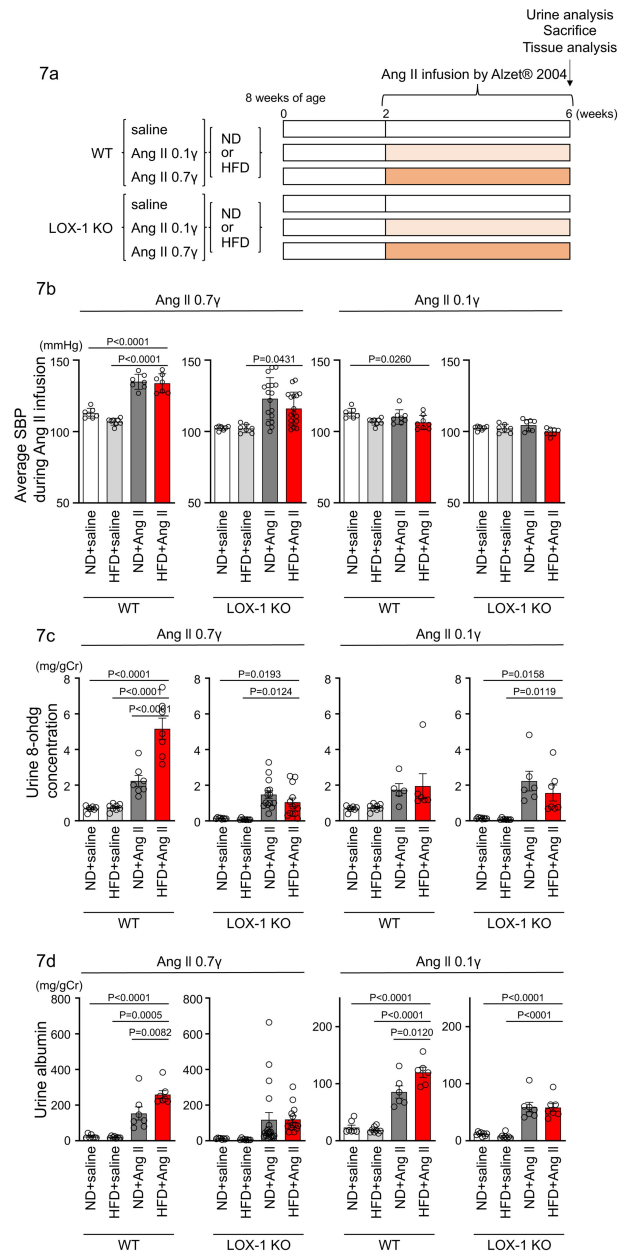


Fig. 7

| Oxidized LDL-inducible diet exacerbates Ang II-induced renal dysfunction in wildtype mice, but not in LOX-1 knockout mice

a A Schematic protocol for the animal experiments. Eight-week-old male WT mice and male LOX-1 KO mice were fed either an ND or an HFD for 6 weeks. After 10 weeks of age, the mice were treated over a 4-week period with infusions of either saline or Ang II. Ang II was administered at two dosage levels: a subpressor dose of 0.1 μ g and a pressor dose of 0.7 μ g, delivered via subcutaneously implanted osmotic pumps. At the end of the infusion period, urine was collected, the animals were sacrificed, and comprehensive tissue analysis was conducted to evaluate the renal effects of the treatments.

b Average systolic blood pressure (SBP) measured at half-week intervals in WT and LOX-1 KO mice during the 4-week infusion period.

c, d Urine 8-OHdG concentrations (mg/g creatinine [Cr]) (c) and urine albumin concentrations (mg/g creatinine [Cr]) (d) in WT and LOX-1 KO mice at the conclusion of the 4-week infusion period.

Data are represented as mean $\pm$ SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-d).

For many of these genes, a synergistic effect of the combination of a pressor dose of Ang II and HFD was not observed in LOX-1 KO mice (**Fig. 8a** [↗](#), **Fig. S8a** [↗](#)). Seven genes displayed discernible differences between HFD- and ND-fed WT mice treated with a subpressor dose of Ang II (**Fig. 8b** [↗](#), **Fig. S8b** [↗](#)). Other than *fibronectin*, no differences were observed between ND- and HFD-fed LOX-1 mice exposed to a subpressor dose of Ang II (**Fig. 8b** [↗](#), **Fig. S8b** [↗](#)). In relation to AT1 and LOX-1 expression, no variations emerged between the ND and HFD groups when subjected to the corresponding dose of Ang II treatment in both WT and LOX-1 KO mice (**Fig. S8a** [↗](#), **b** [↗](#)).

In the histological analysis of kidney samples, in contrast to the gene expression data, the administration of either a subpressor or pressor dose of Ang II, both in isolation and in combination with HFD for 4 weeks, did not reveal any notable increase in fibrosis, as evaluated by Masson-Trichrome staining (**Fig. S9a** [↗](#)). Similarly, there was no significant change in the degree of mesangial expansion or glomerular area, as assessed by PAS staining. (**Fig. S9b** [↗](#)).

Finally, immunofluorescence staining of mouse kidney specimens was performed to detect the colocalization sites of LOX-1 and AT1 in the kidney. As shown in **Fig. 9a** [↗](#) and **Fig. 9b** [↗](#), LOX-1 and AT1a were predominantly colocalized in the renal tubules, but not in the glomeruli. We also performed co-immunofluorescence staining with megalin, a well-established marker of proximal renal tubules, as shown in **Fig. S10** [↗](#). Both AT1 and LOX-1 were observed to colocalize with megalin, especially at the brush borders, indicating their presence within the same renal compartments involved in AT1/LOX-1 signaling.

Discussion

We conducted a series of experiments to provide empirical support for our hypotheses. We propose that the simultaneous binding of Ang II to AT1 and oxLDL to LOX-1 triggers distinct and more pronounced structural modifications in AT1 than the individual modifications induced by each ligand. This structural alteration in AT1 leads to the enhanced activation of Gq signaling.

To confirm AT1 and LOX-1 colocalization and interaction at the cell membrane, our previous study used in situ proximity ligation assays (PLA) and membrane protein immunoprecipitation assays in CHO cells expressing tagged AT1 and LOX-1, successfully demonstrating AT1/LOX-1 complex formation ¹³[↗](#). Additionally, we provided histological evidence of their co-localization with megalin in proximal renal tubules to support these findings, although the limitation remains regarding the lack of direct in vivo evidence for membrane co-localization of LOX-1 and AT1. While additional co-staining with other markers to identify specific cell types was not conducted, the prominent localization of AT1R with megalin in our study provides strong evidence of its expression in proximal renal tubules, consistent with established findings regarding AT1R presence in this nephron segment. Previous studies have documented AT1R expression in various renal cells, including mesangial, interstitial, and juxtaglomerular (JG) cells, as well as proximal tubules. In our immunofluorescence analysis, however, we did not observe significant AT1R expression in the glomerulus or mesangium. The pronounced expression of AT1R in proximal tubules aligns with previous reports ²⁶[↗](#), though limitations in immunofluorescence sensitivity do not exclude AT1R presence in other compartments. Notably, our focus on proximal tubules enabled clear observation of AT1/LOX-1 co-localization, especially under oxLDL and AngII stimulation. This interaction underscores a potential focal point for AT1R/LOX-1 signaling in kidney disease pathogenesis within the renal system.

At the cellular level, live imaging analysis (**Fig. S1** [↗](#)) displayed a limited number of overlapping LOX-1 and AT1R puncta, which could be attributed to the dynamic and transient nature of the LOX-1 and AT1 receptor interaction. As described in our previous study ¹³[↗](#), this interaction is sensitive to buffer conditions, with complex formation occurring under non-reducing but not reducing conditions. This indicates that the LOX-1 and AT1 interaction is mediated by non-covalent

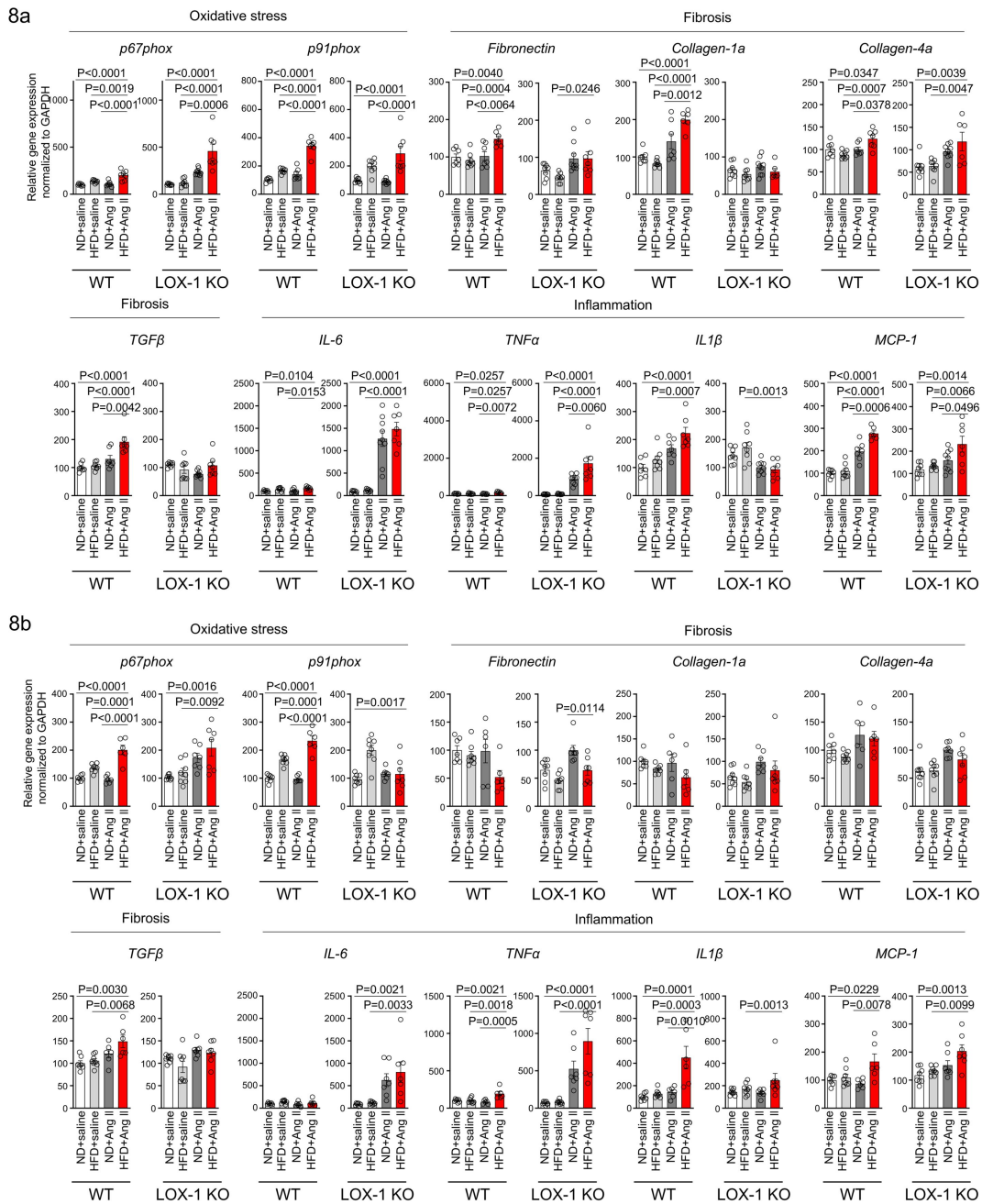


Fig. 8

|A high-fat diet enhanced Ang II-induced renal injury-related gene expression in the kidney in a LOX-1-dependent manner.

Quantitative real-time PCR analysis for gene expression of NADPH components (*p67phox* and *p91phox*), inflammatory cytokines (*IL-6*, *TNFα*, *IL1β*, and *MCP-1*), and fibrosis markers (*TGFβ*, *fibronectin*, *collagen-1a*, and *collagen-4a*) in the kidney harvested from WT and LOX-1 KO mice.

The experimental procedures, including the dietary regimen and Ang II administration, are detailed in [Fig. 7a](#). Mice were administered either a pressor dose of 0.7 μg Ang II (a) or a subpressor dose of 0.1 μg Ang II (b).

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).

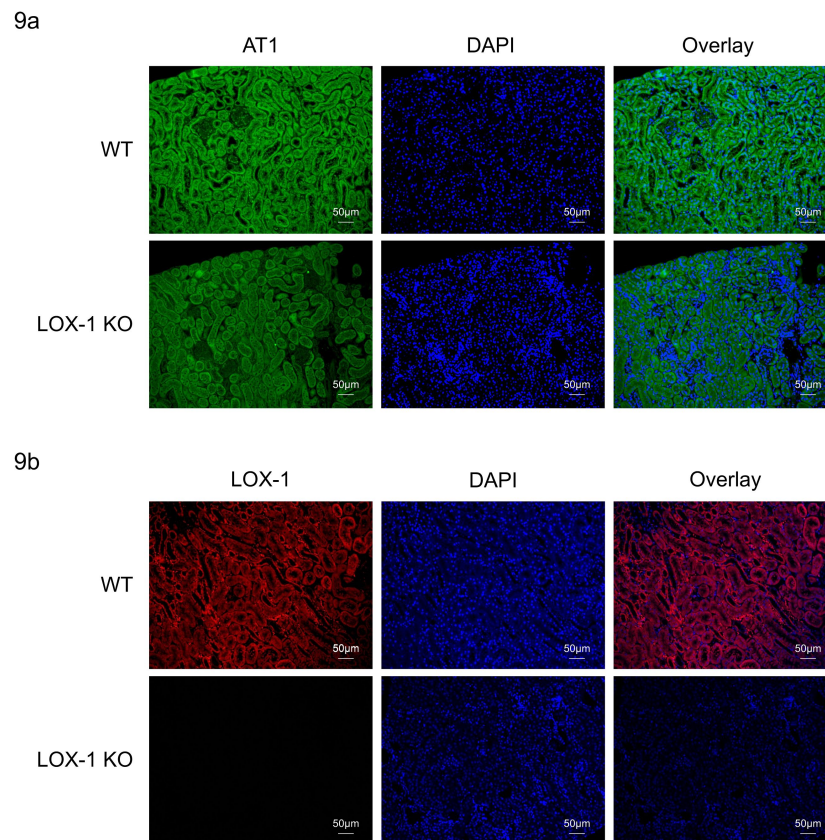


Fig. 9

| LOX-1 and AT1a were predominantly co-localized in renal tubules

a, b Representative images depicting staining for LOX-1 (a) and AT1 (b) in the renal cortex tissues from wildtype mice (WT) and LOX-1 knockout mice (LOX-1 KO). Nuclei are stained blue with DAPI. Green and red signals indicates AT1, while the red signal indicates LOX-1. Overlay images demonstrate the merged visualization of AT1 or LOX-1 with DAPI, highlighting the predominant colocalization of LOX-1 and AT1 in renal tubules as opposed to the glomerulus.

rather than stable covalent bonds, leading these receptors to form and dissociate rapidly. Consequently, only a small fraction of LOX-1 and AT1 receptors may be co-localized at any given time point, explaining the limited number of overlapping puncta observed in live imaging. Importantly, despite this limited spatial overlap, our findings indicate that co-treatment with oxLDL and Ang II significantly enhances Gq signaling. This highlights that the functional impact of the LOX-1/AT1 interaction, especially in response to specific stimuli like oxLDL and Ang II, is more crucial to downstream signaling outcomes than the extent of stable receptor co-localization.

The current experiments showed that oxLDL enhanced the effects of Ang II-induced production of IP1, a downstream signaling molecule associated with Gq activation and subsequent calcium influx, further supporting functional activation of the AT1/LOX-1 complex. However, this effect was observed only in the presence of both LOX-1 and AT1. The differences in activation levels observed between the IP1 assay and the calcium influx assay, both indicators of Gq activity, likely arise due to variations in assay sensitivity. While the absolute differences in the IP1 assay between treatment groups may appear modest, the critical comparison between Ang II alone and Ang II with oxLDL consistently demonstrated significant differences, in alignment with the calcium influx results. Notably, co-treatment with oxLDL reduced the EC50 for IP1 production by 80% compared to Ang II alone, underscoring a robust enhancement of Gq signaling, even though the IP1 assay differences were relatively small in absolute terms.

Considering that oxLDL selectively activates Gi but not Gq through the LOX-1-AT1 dependent pathway¹⁴, it is evident that the observed phenomenon cannot be attributed solely to the additive effect of oxLDL on AT1 activation. Rather, the simultaneous binding of oxLDL and Ang II to their respective receptors, LOX-1 and AT1, which form a single complex, underlies this phenomenon. Indeed, our findings from live imaging of the membrane receptors revealed that the combination of Ang II and oxLDL did not induce any additional influence on the internalization of both receptors upon AT1- β -arrestin activation compared to each ligand alone, suggesting that the quantity of activated receptors is not affected by the combination treatments. Considering the conformation-activation relationship in AT1 activation²⁰, this supports the hypothesis that the simultaneous binding of Ang II and oxLDL in a single AT1-LOX-1 complex induces a greater conformational change, resulting in a more open conformation of AT1 compared with each ligand alone. However, it is technically challenging to directly identify structural modifications of the receptor complex using techniques such as crystal structure analysis. Instead, we utilized AT1 conformational sensors capable of differentiating between the conformational changes induced by Ang II and biased AT1²⁰. Interestingly, we observed that one of the two conformational sensors employed in this study detected an augmented response in the presence of the combination treatment compared with Ang II alone. Importantly, this enhanced response was significantly inhibited when an LOX-1 antibody was introduced, indicating the dependency of LOX-1 on this phenomenon.

These results indicate that the combined treatment with Ang II and oxLDL in the presence of LOX-1 induces a unique conformational change in individual AT1 molecules, which differs from the conformational changes induced by each single treatment alone.

It is important to note that the ability of LOX-1 ligands to enhance Ang II-AT1 signaling is not commonly observed. This was corroborated by the observation that BSA-conjugated AGE, a recognized ligand of LOX-1¹⁷, failed to augment the production of IP1 by Ang II compared to control BSA (**Fig. 1d**). While the exact reason for this discrepancy is not yet understood, it is noteworthy that the predicted particle size of oxLDL²⁷ is significantly larger at 250 Å compared to the maximum particle size of AGE-BSA²⁸, which is 120 Å. This suggests a potentially greater impact of oxLDL on the structural modifications within the AT1-LOX-1 complex. However, further structural analysis is required to validate this hypothesis.

Notably, the Gq bias resulting from combination treatment varied across the mammalian cells examined. Specifically, we observed a combinatorial effect exclusively in renal epithelial and fibroblasts, whereas vascular endothelial and smooth muscle cells did not display the same response. In these renal cells, we observed increased Gq signaling, along with other cellular phenomena such as calcium influx, changes in gene expression, and alterations in cellular characteristics, including myofibroblast activation and cell proliferation. While our study demonstrates a significant upregulation of α -SMA, a well-established marker of myofibroblast, in renal epithelial and fibroblast cells exposed to combined Ang II and oxLDL treatment, we acknowledge the evolving understanding of EMT, particularly the role of partial epithelial-mesenchymal transition (pEMT) in CKD. Notably, pEMT has garnered attention under conditions of inflammation, oxidative stress, and elevated TGF- β , which are also relevant to our Ang II and HFD model. Unlike full EMT, where epithelial cells completely transition into mesenchymal cells, pEMT represents a state in which epithelial cells partially acquire mesenchymal characteristics, such as increased α -SMA expression and the secretion of pro-fibrotic cytokines, while remaining attached to the basement membrane without fully transitioning into fibroblasts³¹. Importantly, previous studies using the unilateral ureteral obstruction (UUO) model suggest that full EMT is unlikely to play a significant role in renal fibrosis and that most kidney fibroblasts are thought to originate from interstitial cells rather than via EMT²⁹. The observed increase in α -SMA in our model may therefore indicate a pEMT-like state, indirectly contributing to fibrosis by promoting cytokine and growth factor release rather than directly driving fibroblast generation. This interpretation aligns with findings from other kidney fibrosis models, including the UUO model, which shares pathophysiological features such as inflammation and oxidative stress with our model. Given these considerations, the increased α -SMA expression observed in our study may be indicative of pEMT rather than definitive evidence of EMT directly contributing to fibroblast differentiation. Additionally, extrapolating *in vitro* pEMT findings to *in vivo* models presents inherent challenges, as detecting these subtle phenotypic changes remains complex³¹. Further mechanistic investigations are required to clarify the contribution of pEMT to renal fibrosis. Nevertheless, our findings support the possibility that pEMT may contribute to fibrosis within the specific pathophysiological context of Ang II and HFD co-administration.

Additionally, fibroblast proliferation, as assessed by the BrdU assay, was notably enhanced by the combined treatment, as opposed to each treatment administered separately. Interestingly, the use of 5 μ g/mL oxLDL in our study showed a tendency to decrease proliferation, which was counteracted by the administration of an ARB but not a Gq inhibitor. These findings suggest that the presence of Ang II leads to a significant transformation in the function of oxLDL, primarily due to its altered influence within the AT1-LOX-1 complex. Taken together, these results suggest that the simultaneous binding of oxLDL to LOX-1 and Ang II to AT1 results in a Gq-biased shift in AT1 activation, leading to a cellular phenomenon that could potentially contribute to renal fibrosis.

In an animal study, we introduced a biological environment in which both circulating Ang II and oxLDL (an LOX-1 ligand) were increased in mice. Previous studies using mice fed an HFD have consistently reported the onset of renal injury, as determined by various measurements^{32–36}. In contrast, 6 weeks of HFD feeding without Ang II treatment did not alter renal function in our mice. This can be attributed to the lack of obesity induced by the HFD in this study. We used this diet based on a previous study that confirmed increased circulating LOX-1 ligand levels without body weight gain in mice³⁷. In our experiments, this diet led to a decrease in body weight, likely due to reduced food intake in HFD-fed mice (**Fig. S6d**, **e**). Although modest, this weight reduction may influence renal function. Obesity is a well-established and clinically proven risk factor of renal dysfunction. The mechanisms underlying this association are complex and involve various factors other than lipid abnormalities, such as hemodynamic changes that affect kidney circulation and the impact of adipose tissue on the production of adipokines and other inflammatory mediators^{38,39}. Consequently, we observed the influence of elevated lipid particle levels on renal function, independent of obesity. We found that the effect of an HFD became obvious with an increase in the Ang II load in WT mice. In particular, in WT mice treated

with high-dose Ang II, which elevated systolic BP by approximately 30 mm Hg, an HFD induced notable increases in urinary reactive oxygen species and urinary albumin. In contrast, an HFD had no impact on Ang II-infused LOX-1 KO mice, as evidenced by equivalent urinalysis measurements for renal injury between the diets. Correspondingly, simultaneous administration of an HFD and Ang II resulted in a consistent alteration in the expression of genes related to renal injury, including fibrosis, inflammation, and oxidative stress, except for some genes in WT mice, but not in LOX-1 KO mice. This strongly suggests that the combined effect of Ang II and HFD on renal function is LOX-1 dependent. Nevertheless, it should be noted that the effect of high-dose Ang II infusion on BP tended to be less pronounced in LOX-1 KO mice compared to WT mice, although there were no differences in BP elevation between the diets in each group of mice. This reduction in BP may contribute to the decreased renal injury observed in LOX-1 KO mice, independent of the AT1/LOX-1 interaction. These findings align with those of previous studies indicating that LOX-1 knockout mice show resistance to Ang II-induced elevation of BP^{25,40,41}. Specifically, when mice were infused with 2 μ g Ang II (equivalent to 25 g mice) for a duration of 28 days, wildtype mice experienced a BP increase exceeding 180 mmHg, while LOX-1 knockout mice demonstrated a reduction of approximately 40 mmHg in this elevation²⁵. Furthermore, the same research group reported that Ang II infusion led to less severe renal injury in LOX-1KO mice compared to WT mice⁴⁰. Importantly, these findings were observed in mice fed with an ND, suggesting that the protective effect of LOX-1 loss-of-function against Ang II-induced elevated BP occurs through the LOX-1-AT1 complex, independent of the presence of oxLDL. Additionally, under suppressor doses of Ang II, where no significant differences in BP were observed, HFD-fed WT mice still exhibited increased renal injury compared to ND-fed mice, an effect that was reduced in LOX-1 KO mice. These findings suggest that the protective effects of loss-of-function of LOX-1 are partly independent of BP changes, underscoring the role of the AT1/LOX-1 interaction in renal injury with Ang II and HFD co-treatment. We recognize that our single time-point analysis (1.5 months post-treatment) also may limit these observations, as the effects of AT1/LOX-1 interaction on renal injury could vary with treatment duration. Taken together, the effects of LOX-1 on AT1 signaling are complex, involving both ligand-dependent and -independent mechanisms, and further investigation is required for a comprehensive understanding of the LOX-1-AT1 interaction.

Regarding the effects of AT1 and LOX-1 interaction on the renin-angiotensin system (RAS) and blood pressure, oxLDL binding to LOX-1 enhances AT1 receptor-mediated Gq signaling, thereby promoting Ang II effects such as vasoconstriction, oxidative stress, and inflammation, all of which contribute to elevated blood pressure. However, in HFD-fed mice treated with a pressor dose of Ang II, plasma aldosterone levels showed an increasing tendency but remained statistically non-significant compared with ND-fed mice (ND: 102.8 ± 11.6 pg/mL vs. HFD: 141.8 ± 15.0 pg/mL, $P = 0.081$), as shown in **Fig S7**, indicating a limited response in aldosterone production under these conditions. Additionally, BP did not significantly change (**Fig S6f**), potentially due to heterogeneous cellular responses across cell types, as indicated by the lack of reaction in vascular endothelial cells, vascular smooth muscle cells, and macrophages (**Fig S2**), and/or possibly due to aldosterone saturation from the high Ang II dose.

Finally, the current findings unequivocally demonstrated the molecular interactions between key molecules associated with dyslipidemia and hypertension in the kidneys. Moreover, this interaction can be effectively inhibited by ARBs, suggesting an additive effect in preventing the development of CKD, particularly in patients with hypertension and dyslipidemia. Interestingly, Ang II-dependent hypertensive animal models, including constriction of the renal artery and infusion of Ang II in rats and mice, have revealed a progressive increase in intrarenal Ang II levels, surpassing what can be accounted for by circulating Ang II levels alone⁴². This is due to Ang II-dependent renal activation of the RAS, as indicated by increased urinary angiotensinogen (AGT) secretion^{42,43}. Importantly, the elevation of urinary AGT is also evident in patients with various pathologies, including hypertension and CKD, implying that renal Ang II levels increase even in individuals who do not exhibit elevated levels of circulating Ang II^{21,43,44}. Taken

together, the current finding of the synergistic effect of Ang II and oxLDL on AT1 activation in renal tissue is highly relevant for the development of kidney disease. RAS inhibitors, ARB, and ACE inhibitors are prioritized therapies to prevent the development of CKD with proteinuria in patients with hypertension⁴⁵. In addition to the well-established inhibitory effects of Ang II on the contraction of efferent arterioles⁴⁶, a novel renal protective action of RAS inhibitors has been proposed. Particularly in hypertension accompanied by dyslipidemia, RAS inhibitors may exhibit anti-inflammatory, antifibrotic, and antioxidant effects in the kidneys by inhibiting the Gq signaling pathway through the AT1-LOX-1 complex in renal tubular cells and fibroblasts. In terms of ARBs' effectiveness in inhibiting AT1/LOX-1 receptor conformational changes, all ARBs generally block the downstream signaling from AT1-LOX-1 interaction by preventing Ang II binding to AT1. Our previous study also showed that ARBs, including olmesartan, telmisartan, valsartan, and losartan, could inhibit AT1 activation by oxLDL in the absence of Ang II¹³. However, certain ARBs—olmesartan, telmisartan, and valsartan—also act as inverse agonists, reducing baseline AT1 activity by preventing conformational changes even without Ang II¹³. This inverse agonist property may offer additional therapeutic benefits by reducing receptor activation beyond what is achieved by simple antagonism in pathological states where AT1 activation occurs independently of Ang II, such as in oxLDL presence. Collectively, the current findings suggest that RAS inhibitors, some of which possess inverse agonist properties, can concomitantly mitigate the effect of increased renal Ang II and oxLDL levels on the development of CKD in patients with hypertension and dyslipidemia, although direct evidence in clinical studies to support this remains to be elucidated.

This study has several limitations as follows: 1) The kidney, a complex organ vital for maintaining homeostasis, comprises a myriad of distinct cell types working in concert to execute its multifaceted functions⁴⁷. The experiments conducted in mice raised questions regarding the specific cell types implicated in the synergistic effect of Ang II and oxLDL within the LOX-1-AT1 complex in the kidney. Addressing this issue would ideally require single-cell analysis, which is a challenge for future research. 2) The study employed systemic LOX-1 knockout mice. For a more detailed analysis, a phenotypic investigation using renal tissue-specific LOX-1 and AT1 knockout mice is required. Of particular importance is the analysis using tubule-specific knockout mice, in which the localization of these components has been verified through immunohistochemical staining. 3) The administration of Ang II (both pressor and subpressor doses) and its combination with HFD did not result in any histological changes in terms of fibrosis and mesangial expansion, despite the observed alterations in the associated gene expression and urinalysis for renal injury. Alterations in gene expression and urinalysis for renal injury may be sensitive and relatively early phenomena, and this discrepancy could potentially be attributed to the relatively short intervention duration of 4 weeks for combined HFD and Ang II administration. Therefore, concurrent pathohistological alterations in the renal tissue might become evident with a more extended intervention period. 4) This study was limited by its inability to detect the amplification of Gq signaling in mouse renal tissue due to the concurrent administration of Ang II and HFD. Overcoming this limitation is a challenge for future studies.

In conclusion, the current findings suggest that the simultaneous binding of oxLDL and Ang II to their respective receptors within the complex induces a distinct conformational change compared with the effect of each ligand alone. This unique conformational change results in the heightened activation of G protein signaling and subsequent unfavorable cellular reactions in renal component cells. The relevance of this phenomenon was confirmed in mouse models, in which renal dysfunction was prominently exacerbated when there was a concomitant increase in Ang II and oxLDL levels. Notably, this effect was abolished by the deletion of LOX-1, indicating LOX-1 dependency of this *in vivo* phenomenon (Fig. 10). These findings indicate the clinical relevance of the direct interaction between hypertension and dyslipidemia and further support the clinical significance of RA inhibition in treating patients with CKD.

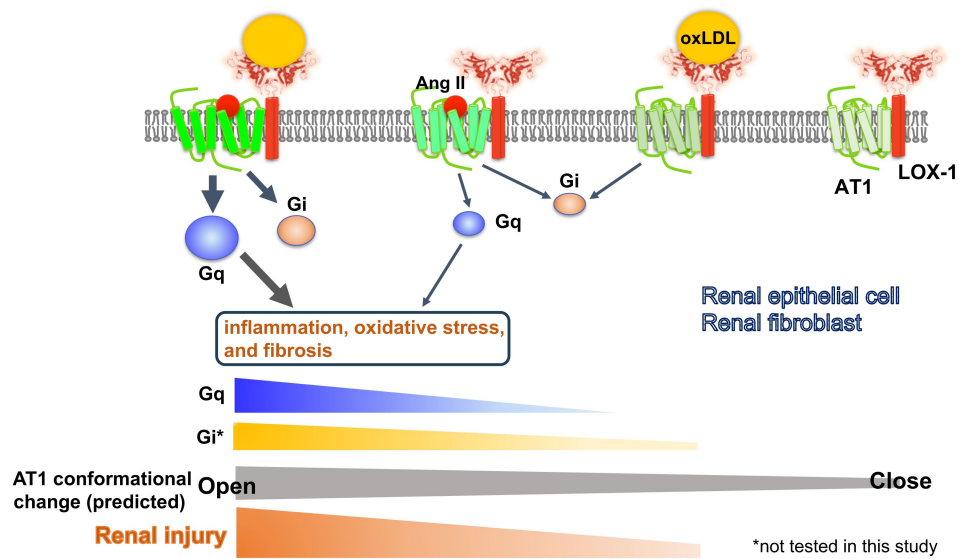


Fig. 10

| Schematic overview of the AT1 and LOX-1 Interaction dynamics in renal cells

This schematic summary illustrates the predicted structure-activation relationship of the AT1 receptor within the LOX-1-AT1 complex in renal component cells. This highlights how the simultaneous binding of Ang II to AT1 and oxLDL to LOX-1 induces conformational changes in AT1. These changes were more pronounced than those triggered by the individual ligands. Such structural alterations have been proposed to amplify Gq signaling pathway activation, subsequently leading to renal damage.

Materials and methods

Cell culture and materials

HUVECs and BAECs were cultured in EGM-2 (Lonza, Basel, Switzerland). Cells with fewer than five passages were used in the experiments. Transgenic CHO cells were maintained in an F-12 Nutrient Mixture with Glutamax™-I (Thermo Fisher Scientific, MA, USA), 10% fetal bovine serum (FBS; Gibco, USA), and appropriate selection reagents, as described below. CHO-K1 cells were maintained in F-12 Nutrient Mixture with Glutamax™-I and 10% FBS. HAVSMCs were cultured in Dulbecco's modified Eagle's medium/F12 (DMEM/F12) (Nacalai Tesque, Japan) supplemented with 1% penicillin-streptomycin (Fujifilm, Japan) and 10% FBS. A10 cells were grown in DMEM (Wako, Osaka, Japan) supplemented with 10% FBS and 1% penicillin-streptomycin. NRK52E and NRK49F cells (ECACC, UK) were cultured in DMEM (Wako, Japan) supplemented with 5% FBS and the appropriate selection reagents. Gene transcription in CHO cells was induced by adding 100 ng/mL doxycycline (Merck KGaA, Darmstadt, Germany). Cells were incubated at 37°C in 5% CO₂ and 95% air.

Construction of plasmid vectors

For stable transformants, pTRE2hyg vector (Clontech, USA) encoding mutated hAT1 with impaired ability to activate β -arrestin (pTRE2hyg-HA-FLAG-hAT1mg) were created using site direct mutagenesis as previously described.¹⁴ For real-time imaging, LOX-1 tagged with V5-6×His at the C-terminus (V5-LOX-1) was subcloned into pmScarlet_C1 (plasmid #85042; Addgene) (mScarlet-LOX-1). HA-FLAG-hAT1 was subcloned into pcDNA3-EGFP (plasmid #85042; Addgene) (AT1-eGFP).¹⁴

Stable transformants

We constructed CHO-K1 cells expressing tetracycline-inducible human LOX-1 tagged with V5-6×His at the C-terminus (CHO-LOX-1), human HA-FLAG-hAT1 (CHO-AT1), or cells expressing both human LOX-1 and AT1 (CHO-LOX-1-AT1), as previously described.^{13,14} To establish cells expressing both LOX-1 and mutated AT1 (pTRE2hyg-HA-FLAG-hAT1mg), they were co-transfected with the pSV2bsr vector (Funakoshi, Japan) into CHO-LOX-1 using the Lipofectamine 2000 transfection reagent (Thermo Fisher Scientific, USA). The stable transformants were selected with 400 μ g/mL of hygromycin B (Wako, Osaka, Japan) and 10 μ g/mL of blasticidin S (Funakoshi, Japan). The resistant clone expressing LOX-1 and mutated AT1 in response to doxycycline (Calbiochem, USA) was selected for use in the experiments (CHO-LOX-1-AT1mg).¹⁴

Small interfering RNA

NRK52 and NRK47 cells were plated at 50% confluence on the day of transfection. Silencer Select small interfering RNA (siRNAs) for *LOX-1* and *AT1a* (Thermo Fisher Scientific, MA, USA) were transfected into cells in medium without serum or antibiotics using Lipofectamine RNAiMAX (Thermo Fisher Scientific, MA, USA), according to the manufacturer's instructions.

Preparation of oxLDL

Human plasma LDL (1.019–1.063 g/mL), isolated by sequential ultracentrifugation, was oxidized using 20 μ M CuSO₄ in PBS at 37 °C for 24 h. Oxidation was terminated by adding excess EDTA. LDL oxidation was analyzed by agarose gel electrophoresis for migration versus LDL.¹³

Quantification of cellular IP1 accumulation

Gq-dependent activation of phospholipase C was quantified by measuring IP1 using the IP-One assay kit (Cisbio, France) as previously described¹⁹. Briefly, cells were seeded at 80,000 cells/well in 96 well transparent cell culture plates and incubated under serum-free conditions for 24 h. Thereafter, cells were treated for 1 h with IP1 stimulation buffer, including vehicle, native LDL, oxLDL, Ang II, AGE, PTX (Merck KGaA, Darmstadt, Germany), YM-254890 (Fujifilm Wako, Osaka, Japan), and RKI-1448 (Selleck, USA), as described in the text. Cell lysates with Triton X at a final concentration of 1% were transferred to a 384-well white plate, and IP1 levels were measured by incubating the cell lysates with FRET reagents (cryptate-labeled anti-IP1 antibody and d2-labeled IP1 analog).

Quantitative real-time PCR

RNA samples were purified using RNeasy Mini Kit (Qiagen, Germantown, MD, USA). One microgram of RNA was converted into cDNA using a ReverTra Ace qPCR RT kit (TOYOBO, Osaka, Japan) according to the manufacturer's instructions. All genes were evaluated using the ViiA7 Real-Time PCR System (Applied Biosystems, Thermo Fisher Scientific, Waltham, MA, USA). The data were analyzed using the $\Delta\Delta C_t$ method with normalization against the GAPDH RNA expression in each sample. The primer sequences are listed in the Supplemental Table.

Western blotting

Proteins were separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis and electrophoretically transferred onto polyvinylidene fluoride membranes. The membranes were blocked with 5% nonfat dried milk and incubated with primary antibodies overnight at 4 °C. The primary antibodies used in this study were as follows: anti-SMA antibody (1:1000), anti- α -Tubulin antibody (1:1000) (Cell Signaling Technology, Inc, Danvers, MA, USA). The bands were visualized with a chemiluminescence detection system (LAS-4000 mini; GE Healthcare Life Sciences, Buckinghamshire, UK) using Chemi-Lumi One Super (Nacalai Tesque, Kyoto, Japan).

Calcium influx assay

Calcium influx was measured using Fura 2-AM (Dojindo, Kumamoto, Japan) with slight modifications to the manufacturer's protocol. In brief, cells plated in 96 wells were incubated with 5 μ M Fura 2-AM in HEPES buffer saline (20 mM HEPES, 115 mM NaCl, 5.4 mM KCl, 0.8 mM MgCl₂, 1.8 mM CaCl₂, 13.8 mM glucose, pH 7.4) for 1 h at 37°C, followed by replacement with recording medium without Fura 2-AM. Cells were treated with oxLDL, Ang II, or a combination of both at the indicated concentrations. Changes in F340/F380, an index of intracellular calcium concentration, were measured by dual-excitation microfluorometry using a digital image analyzer (Aquacosmos; Hamamatsu Photonics, Hamamatsu, Japan).

BrdU assay

Proliferative activity was assessed using a BrdU Cell Proliferation ELISA Kit (Funakoshi, Japan). NRK49F cells were seeded in 96-well tissue culture plates and incubated with the test reagents for 24 h. After incubating the cells with BrdU, we fixed the cells and denatured their DNA using a Fixing Solution. The plate was washed thrice with Wash Buffer before adding the Detector Antibody. Next, 100 μ L/well of anti-BrdU monoclonal Detector Antibody was added, and the plate was incubated for 1 h at room temperature. Subsequently, 100 μ L/well of Goat Anti-Mouse IgG Conjugate was pipetted and incubated for 30 minutes at room temperature. After five washes, the reaction was stopped by adding Stop Solution to each well. The color of the positive wells changed from blue to bright yellow. Finally, the plate was read at a wavelength of 450/550 nm using a spectrophotometric microtiter plate reader.

Creation of lentivirus encoding AT1 conformational sensors

For lentivirus encoding AT1 conformational sensors, rat AT1 and RlucII were subcloned into the pLVSIIN-CMV Neo vector (Takara Bio, Japan). Next, the FAsH binding sequence (CCPGCC) was inserted between residues K135 and S136 in the third intracellular loop (AT1-ICL3P3), as well as between residues K333 and M334 in the cytoplasmic-terminal tail (AT1-Ctail), utilizing the KOD-Plus Mutagenesis Kit (Toyobo, Japan), at the same site as previously documented¹⁴.

Intramolecular FAsH Bioluminescence resonance energy transfer assay to detect AT1 conformational change

CHO-LOX-1 cells were initially plated onto a white clear-bottom 96-well culture plate at a density of 1×10^5 cells/well. The following day, the cells were transduced with lenti-virus encoding AT1-Ctail or AT1-ICL3P3²⁰ in 10% FBS. After 24 h of transduction, the cultures were transferred to serum-free conditions and incubated for an additional 24 h. Add 1.5 μ M FAsH-EDT₂ labeling reagent of TC-FAsH™ II In-Cell Tetracysteine Tag Detection Kit (Thermo Fisher Scientific, MA, USA), wash twice with 250 μ M BAL buffer, and assays were promptly conducted on a Spark® microplate reader (TECAN, Switzerland). The BRET ratio (emission mVenus/emission Rluc) was calculated as follows: Following a 3-minute baseline reading (with the final baseline reading presented at 0), cells were exposed to vehicle, oxLDL alone, AII alone, a combination of AII and oxLDL, or a combination of AngII, oxLDL, and LOX-1 antibodies. The BRET ratios were calculated every 16 seconds for a total of 320 seconds and the relative change in intramolecular BRET ratio was calculated by subtracting the average BRET ratio measured for cells stimulated with vehicle at each time point.

Analysis of LOX-1 and AT1 dynamics by real-time imaging

Live imaging was performed using previously reported methods¹⁹. Briefly, twenty-four hours prior to imaging experiments, CHO-K1 cells were transfected with LOX-1-mScarlet and AT1-eGFP by electroporation. Subsequently, the cells were seeded in a 35-mm glass base dish (Iwaki, Japan) that had been pre-coated with a 1000X diluted solution of 10 mg/mL poly-L-lysine (ScienCell, USA) one hour before seeding. The growth medium was substituted with imaging buffer (pH 7.4), which consisted of 125 mM NaCl, 5 mM KCl, 1.2 mM MgCl₂, 1.3 mM CaCl₂, 25 mM HEPES, and 3 mM D-glucose, with the pH adjusted to 7.4 using NaOH. Dynamic images of the cells were acquired at 25 °C using a SpinSR10 inverted spinning disk-type confocal super-resolution microscope (Olympus, Japan). The microscope was equipped with a 100x NA1.49 objective lens (UAPON100XOTIRF, Olympus, Japan) and an ORCA-Flash 4.0 V2 scientific CMOS camera (Hamamatsu Photonics KK, Japan) at 5-second intervals. The imaging experiment was conducted using CellSens Dimension 1.11 software, employing a 3D deconvolution algorithm (Olympus, Japan), and the number of puncta was determined using ImageJ1.53K¹⁹.

Animals and diets

Male WT mice (C57BL/6J) and LOX-1 KO mice with a C57BL/6 background were used in this study. LOX-1 KO mice were generated as described previously⁴⁸. Mice were housed in a temperature-controlled (20-22°C) room on a 12 hours light/dark cycle and fed an ND (MF; Oriental Yeast, Osaka, Japan) or an HFD (High Fat Diet without DL- α -tocopherol, CLEA Japan Inc, Tokyo, Japan), which reported to increase plasma LOX-1 ligand in ApoE KO mice²¹. All study protocols were approved by the Animal Care and Use Committee of Osaka University and were conducted according to the guidelines of the NIH for the Care and Use of Laboratory Animals.

Blood pressure measurement in mice

The blood pressure of the mice was measured using the tail-cuff method with BP-98A (Softron, Japan). The measurements were performed after restraining the mice. The blood pressure was calculated as the average of 6 readings for each animal at each time point.

Urine tests in mice

Urine tests in mice included the measurement of urine 8-OHdG, creatinine, and albumin concentrations. The DNA Damage (8-OHdG) ELISA Kit (StressMarq Bioscience, Canada), Creatinine Kit L type Wako (Fujifilm, Japan), and Mouse Albumin ELISA Kit (Bethyl Laboratories, Inc., TX, USA) were utilized for these measurements, following their respective instructions.

Plasma LOX-1 ligand concentration

Measurement of LOX-1 ligands containing apoB (LAB) in mouse plasma was performed using a modified protocol based on a previously reported method ³⁷. Briefly, recombinant human LOX-1 (0.25 µg/well) was immobilized on 384 well plate (greiner, Frickenhausen, Germany) by incubating overnight at 4°C in 50 µl of PBS. After three washes with PBS, 80 µl of 20% (v/v) ImmunoBlock (KAC, Kyoto, Japan) was added, and the plates were incubated for 2 h at 25°C. After three washes with PBS, the plates were incubated for 2 h at 25°C with 40 µl of standard oxidized LDL or samples. Samples were prepared by 4-fold dilution of plasma with HEPES-EDTA buffer (10 mM HEPES, 150 mM NaCl, 2 mM EDTA, pH 7.4), and standards were prepared by dilution of oxidized LDL with HEPES-EDTA buffer. Following three washes with PBS, the plates were incubated for 1 h at 25°C with chicken monoclonal anti-apoB antibody (HUC20, 0.5 µg/mL) in HEPES-EDTA containing 1% (w/v) BSA. After three washes with PBS, the plates were incubated for 1 h at 25°C with peroxidase-conjugated donkey anti-chicken IgY (Merck, NJ, USA) diluted 5000 times with HEPES-EDTA containing 1% (w/v) BSA. After five washes with PBS, the substrate solution containing 3,3',5,5'-tetramethylbenzidine (TMB solution, Bio-Rad Laboratories, CA, USA) was added to the plates and incubated them for 30 min at room temperature. The reaction was terminated with 2 M sulfuric acid. Peroxidase activity was determined by measuring absorbance at 450 nm using a SpectraMax 340PC384 Microplate Reader (Molecular Devices, CA, USA).

Tissue preparation

Kidneys were perfused with cold PBS before removal. Kidney samples were rapidly excised. A quarter of samples were stored at 4°C in RNAlater (Thermo Fisher Scientific, MA, USA) for RNA extraction. The remaining quarters were fixed in 4% paraformaldehyde overnight at 4°C for histological evaluation. The remaining half was snap-frozen in liquid nitrogen and stored at -80°C for further analysis.

Periodic acid-Schiff and Masson-Trichrome staining

The degree of glomerular mesangial expansion and glomerular area (representing the structural integrity of the glomeruli) were assessed in a blinded manner using periodic acid-Schiff (PAS) staining. Collagen accumulation was determined by Masson-Trichrome (MTC) staining. For MTC staining, the area displaying fibrosis was quantitatively evaluated in a blinded manner by measuring the blue staining in six strongly magnified fields of view using the ImageJ software, and the average was calculated after determining the ratio of the total area.

Fluorescent immunostaining

For fluorescent immunostaining of LOX-1, AT-1, and meglin, the mice were perfused with cold saline before tissue removal. After 3 days of zinc fixation, the tissue was replaced with 70% ethanol. The 3-µm-thick kidney tissue sections were immunohistochemically stained with antibodies against ATGR (1:200, Cosmo Bio, Japan), OLR-1 (1:200, TS58 from the laboratory of T.S.,

Shinshu University School of Medicine, Nagano, Japan), and megalin (1:200, BiCell Scientific, MO, USA). Following deparaffinization (using Lemosol and gradient ethanol) and rehydration, the slices were subjected to antigen retrieval by autoclaving in citrate buffer (0.01 M; pH 6.0). Subsequently, the slices were washed thrice with PBS and blocked with 5% bovine serum albumin for 30 min at room temperature. The slides were then incubated with primary antibodies for 2 h at room temperature. Goat anti-Rabbit IgG (H+L) High Cross-Adsorbed Secondary Antibody, Alexa Fluor Plus 488 and 594 (Thermo Fisher Scientific, MA, USA) were used as secondary antibodies for ATGR and OLR-1, respectively. After incubation with secondary antibodies for 1 h at room temperature, the slices were washed with PBS. Finally, slides were sealed and photographed. Visual analyses were performed using a BZ-800L microscope (Keyence, Japan).

Statistical analyses

All data are presented as the mean $\pm$ SEM. Differences between two treatments or among multiple treatments were determined using the Student's t-test or one-way ANOVA followed by Tukey's multiple comparison test.

Supplemental Figure Legends

Fig. S1

| Live-imaging analysis of membrane LOX-1 and AT1 in response to the co-treatment of oxLDL with AngII.

Real-time membrane imaging of CHO cells co-transfected with LOX-1-mScarlet and AT1-eGFP in response to oxLDL (10µg/ml) in the combination of Ang II (10⁻⁷M) (Supplemental Video). A count of puncta was performed using separate images visualizing LOX-1-mScarlet (red puncta) and AT1-eGFP (green puncta) immediately before and 3 min after ligand application.

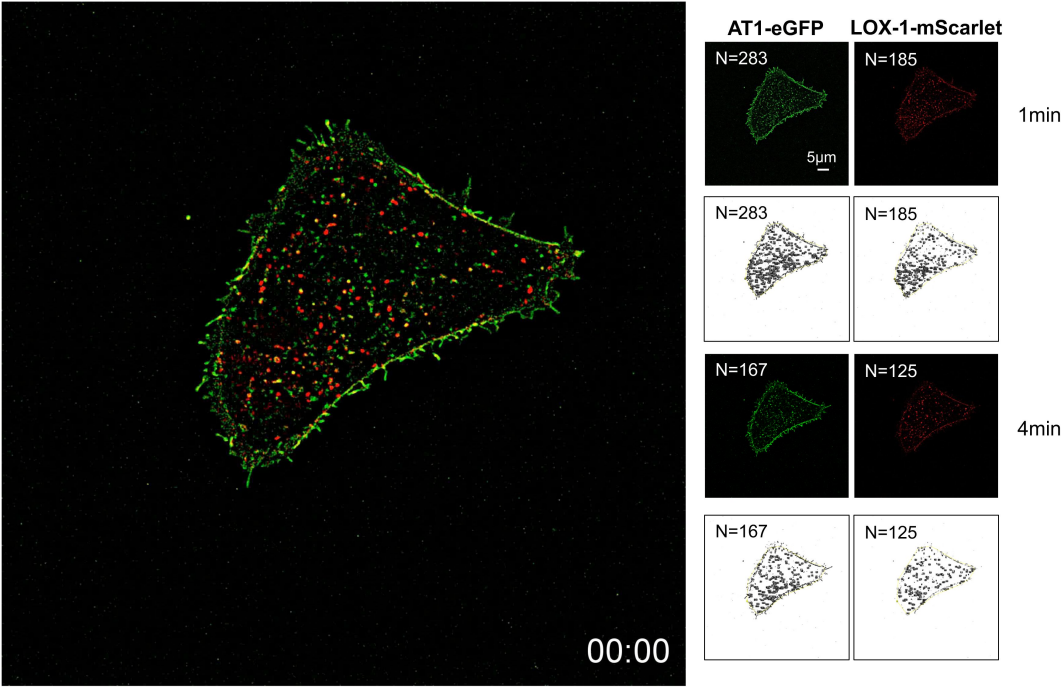


Fig. S2

| Oxidized LDL in combination with Ang II do not increase cellular IP1 content in human umbilical vein endothelial cells and bovine vascular endothelial cells, human aortic vascular smooth muscle cells, and rat macrophages

IP1 concentration in response to oxLDL (10µg/ml) in the combination of Ang II (10⁻⁷M) in HUVECs (human umbilical vein endothelial cell), BAECs (bovine aortic endothelial cell), HAVSMCs (human aortic vascular smooth muscle cell), and A10 cells (rat macrophages).

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test (n=5 for each group).

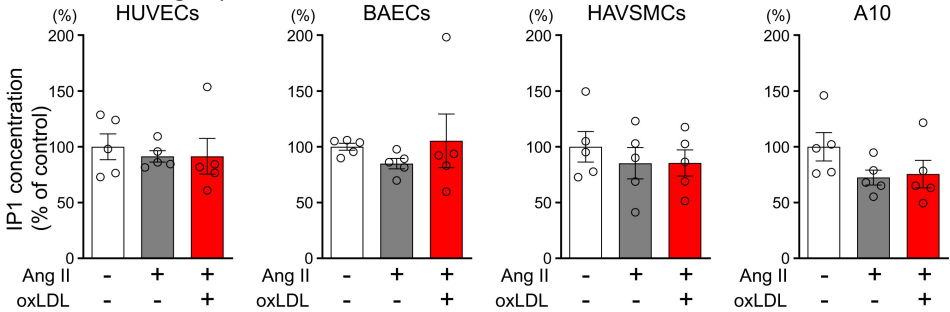


Fig. S3

| Efficiency of siRNA-mediated knockdown for *AT1a* and *LOX-1* in NRK52E and NRK49F cells

NRK52E and NRK49F cells were transfected with siRNAs against scrambled siRNA, *AT1a* or *LOX-1*. The efficiency of siRNA-mediated gene silencing was quantified by assessing *AT1a* and *LOX-1* expression levels using quantitative real-time PCR. Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test.

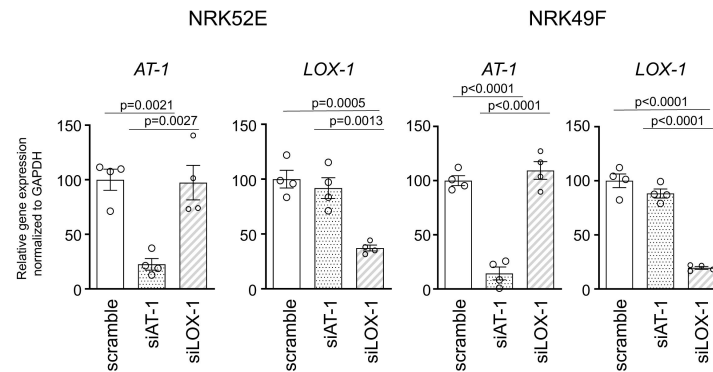
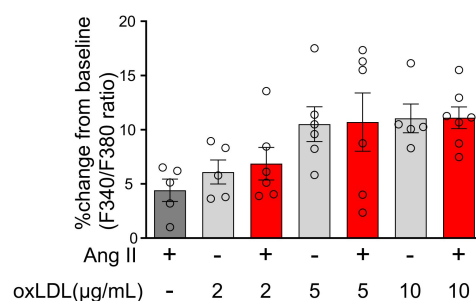


Fig. S4

| Calcium influx was not induced by either the combination treatment of Ang II or oxLDL or each treatment alone in NRK52E cells

Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL at the concentrations detailed in the figure for NRK49E cells (n=5-7 for each group).

Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test.



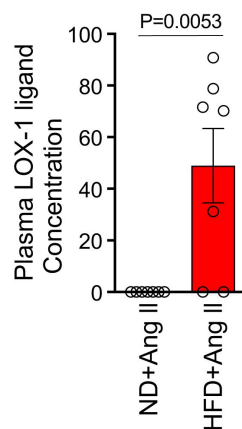


Fig. S5

| High Fat Diet used in the study prominently increased plasma LOX-1 ligand concentration

Plasma LOX-1 ligand concentration of 14-week-old wildtype mice upon ND and HFD for 6 weeks. From 10 weeks of age, these mice also received concurrent 4-week infusions of Ang II at a pressor dose of 0.7 μ g, administered through subcutaneously implanted osmotic pumps.

Data are represented as mean $\pm$ SEM. Differences were determined using Student's t-test (n=7 for each group).

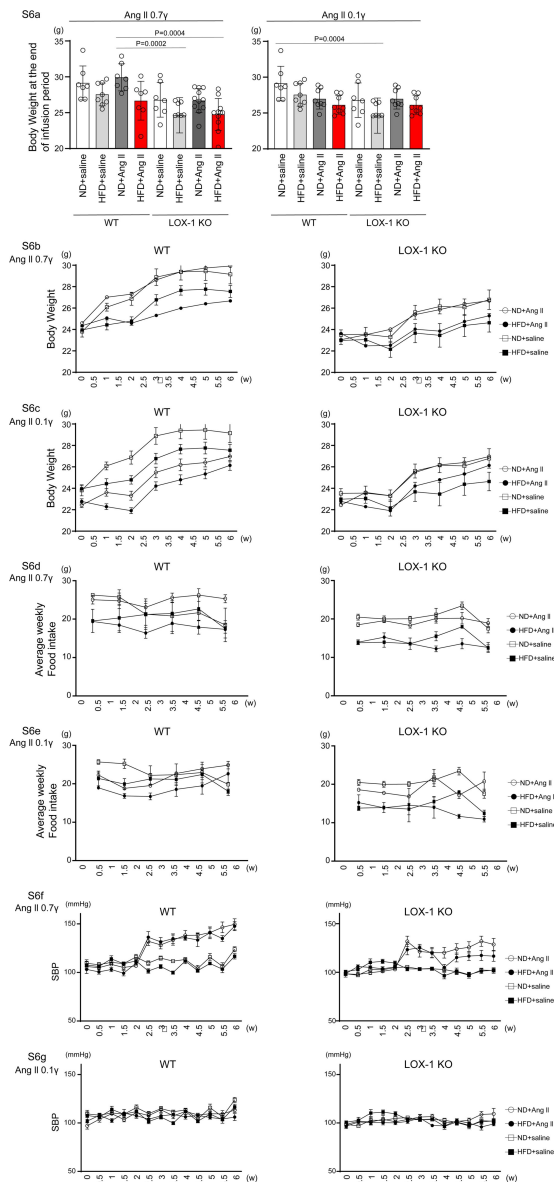


Fig. S6

Impact of Diet and Ang II Infusion on Body Weight and Systolic Blood Pressure in Mice

a Final Body weight: This figure shows the body weights of WT and LOX-1 KO mice at the end of the 4-week infusion period, highlighting the effects of diet and pharmacological treatment.

b, c Serial Body Weight Changes: These graphs depict the progression of body weight over time in WT and LOX-1 KO mice, illustrating the impact of the dietary regimen and Ang II infusion on weight dynamics.

d, e Food Intake: Food intake Trajectory: Serial measurements of weekly food intake (g/week per mouse) are presented for WT and LOX-1 KO mice. The figures show the changes in food intake over the course of the study, corresponding to the administration of a pressor dose of 0.7 γ (d) and a subpressor dose of 0.1 γ (e) of Ang II, respectively.

f, g Systolic Blood Pressure Trajectory: Serial measurements of systolic blood pressure (SBP) obtained using the tail-cuff method are presented for WT and LOX-1 KO mice. The figures show the changes in SBP over the course of the study, corresponding to the administration of a pressor dose of 0.7 γ (f) and a subpressor dose of 0.1 γ (g) of Ang II, respectively. Beginning at 8 weeks of age, mice were fed either an ND or an HFD for 6 weeks. From 10 weeks of age, coinciding with the 2-week time point in the figure, the mice underwent a 4-week period of infusion with either vehicle or Ang II. The infusion was delivered at specific dosage levels through subcutaneously implanted osmotic pumps. Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a).

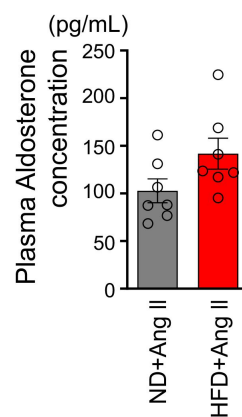


Fig. S7

| No significant difference was found in plasma aldosterone concentration between a normal diet and a high fat diet-fed wildtype mice with a pressor dose of Ang II

Plasma aldosterone concentration in 14-week-old wild-type mice fed an ND or an HFD for 6 weeks. From 10 weeks of age, these mice also received concurrent 4-week infusions of Ang II at a pressor dose of 0.7 μ g, administered through subcutaneously implanted osmotic pumps.

Data are represented as mean $\pm$ SEM. Differences were determined using Student's t-test (n=7 for each group).

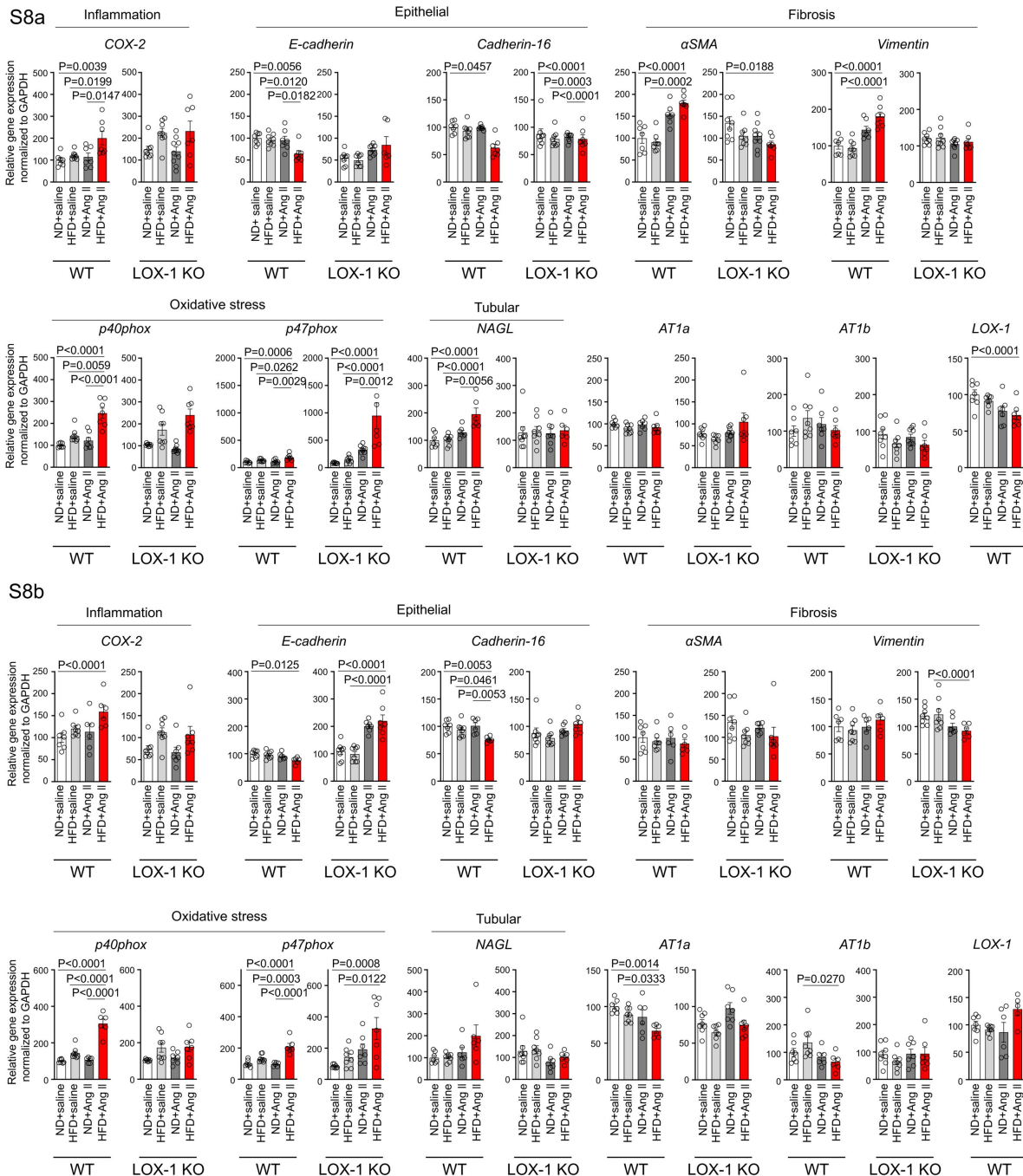


Fig. S8

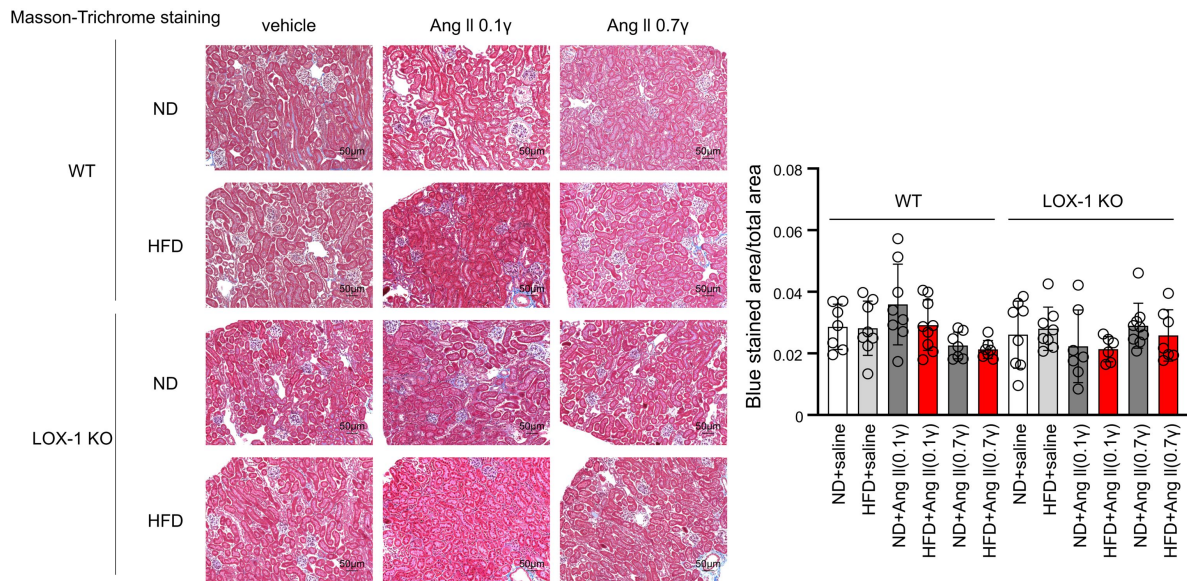
|A high-fat diet enhanced Ang II-induced renal injury-related gene expression in the kidney in a LOX-1-dependent manner.

Quantitative real-time PCR analysis for gene expression of NADPH components (*p40phox* and *p47phox*), inflammatory gene (*COX-2*), fibrosis markers (*αSMA* and *vimentin*), epithelial markers (*E-cadherin* and *cadherin-16*), tubular marker (*NAGL*), *AT1a*, *AT1b*, and *LOX-1* in the kidney harvested from WT and LOX-1 KO mice.

The experimental procedures, including the dietary regimen and Ang II administration, are detailed in **Fig. 7a**. Mice were administered either a pressor dose of 0.7 γ Ang II (a) or a subpressor dose of 0.1 γ Ang II (b).

Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).

S9a



S9b

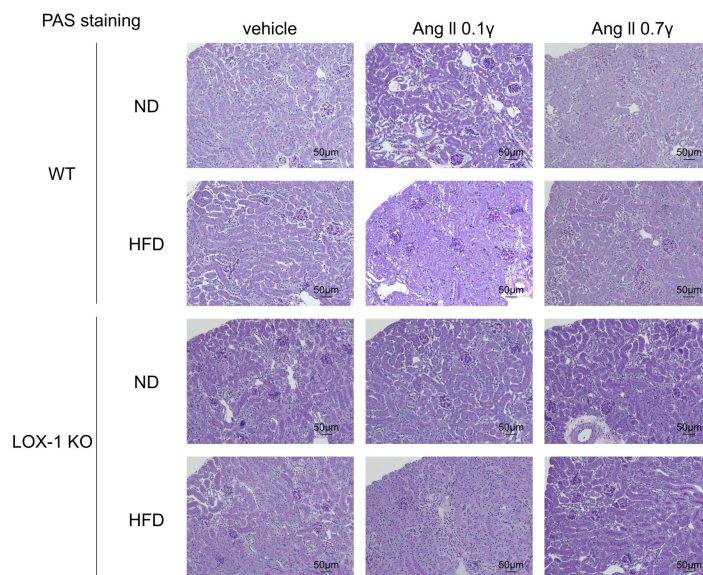


Fig. S9

| The treatment with Ang II, a High Fat Diet, or their combination for 4 weeks did not induce any histological changes indicative of renal injury

a Left: Representative histological images of Masson-Trichrome staining used to detect fibrosis in renal tissues harvested from WT and LOX-1 KO mice. Right: Quantitative analysis by Masson-Trichrome staining. Data are represented as mean $\pm$ SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test.

b Representative histological images of renal tissues harvested from WT and LOX-1 KO mice stained for PAS to assess mesangial expansion and glomerular area, providing insight into the structural integrity of the glomeruli. Eight-week-old mice were fed an ND or an HFD for 6 weeks. From 10 weeks of age, these mice were concurrently treated for 4 weeks with infusions of vehicle or Ang II (a suppressor dose of 0.1 γ or a pressor dose of 0.7 γ) through subcutaneously implanted osmotic pumps.

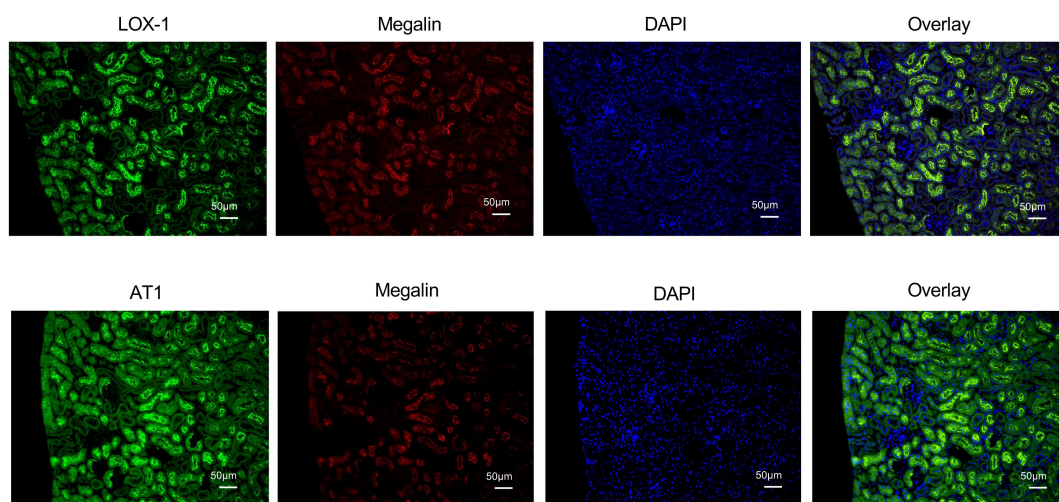


Fig. S10

| LOX-1 and AT1a were co-localized with megalin

Representative images showing co-staining of LOX-1 (upper panels) and AT1 (lower panels) with megalin in renal cortex tissues from WT mice. Nuclei are stained blue with DAPI. Green signals indicate LOX-1 or AT1, while red signals indicate megalin. The overlay images show the merged visualization of AT1 or LOX-1 with megalin and DAPI, highlighting the predominant co-localization of LOX-1 and AT1 at the brush borders.

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Additional information

Author contributions

J.I., Y.H., Y.T., Y.N., T.T., and K.Y. designed the study. J.I. and Y.H. performed most of the experiments. A.K. measured plasma LOX-1 ligand concentrations in mice and assisted with data analysis. J.I. and S.S. performed histological analysis of the renal tissue of mice. J.I., Y.H., Y.T., Y.N., T.T., and K.Y. wrote the manuscript and analyzed the data. C.W., Z.W., G.Y., L.W., Y.N., R.O., T.F., H.K., K.A., H.T., S.S., and K.I. analyzed the data and contributed to the discussion. A.K., Y.I., H.R., and T.S. contributed to the discussion and reviewed and edited the manuscript. All authors contributed to the editing of the final manuscript. Y.T., Y.N., and T.T. had full access to all data in the study and took responsibility for the integrity of the data and accuracy of the data analysis.

Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided in this paper.

Additional files

Supplemental Table [↗](#)

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Reviewer #1 (Public review):

Summary:

In the present study, Dr. Ihara demonstrated a key role of oxLDL in enhancing Ang II-induced Gq signaling by promoting the AT1/LOX1 receptor complex formation.

Strengths:

This study is very exciting and the work is also very detailed, especially regarding the mechanism of LOX1-AT1 receptor interaction and its impact on oxidative stress, fibrosis and inflammation.

Weaknesses:

The direct evidence for the interaction between AT1 and LOX1 receptors in cell membrane localization is relatively weak.

<https://doi.org/10.7554/eLife.98766.2.sa2>

Reviewer #2 (Public review):

While the findings might be valid, there is enough uncertainty that these results should not be considered anything other than preliminary, warranting a more thorough and rigorous investigation.

Comments on revisions:

As the author mentioned that due to the receptor internalisation of AT1 and/or LOX1 induced by AngII or Ox-LDL makes it difficult to detect receptor interaction at the membrane by Co-IP. If so, the GPCR internalisation related pathway should be activated, such as GRKs, arrestin2 could be activated and enhanced during this process, whether they could further provide the evidence for these changes in different groups by Western blot or IF images.

If the authors don't know why the results across experiments can vary so greatly nor control them, how do we know that their interpretation of the very modest intra-experimental variability they observe is correct? They explain away the difference in biosensor activity response to the likely respective insertion sites that were used. While this can be true, and even might be true, it is important to note that the publication they cite shows that the sensors in the third loop and the C-terminus respond very similarly. In fact, the authors concluded: "Our results also suggest that positioning conformational biosensors into ICL3 and the C-tail effectively reports canonical G protein-mediated signaling downstream of the AT1R." Moreover, it is unclear why the less sensitive biosensor (as least as measured by degree of DBRET) is the one that appears to show enhancement. I suppose one could argue that the activity is maximal using the C-tail and one must use a less responsive reporter to detect the effect, but this is a rationalization for an unexplained result rather than a validated mechanistic explanation. If the other results were more compelling, perhaps this would be less of an issue. Finally, they did not explain why a control, non-specific antibody wasn't used for the studies presented in panel 2d. This would have been an easy study to have done in the interim. It also would have been important to test the effect of the LOX1-ab on the effects of AngII treatment alone.

In their response to the gene expression studies, the authors attribute the lack of a robust response for some genes to the low dose of oxLDL that was used but give no justification for their choice for this low dose. More importantly, they present the data for a number of hand-picked genes rather than a global assessment of response. Their justification---cost constraints---isn't sufficient to justify this incomplete analysis. Their selective rt-PCR results are a pilot study.

There is no direct evidence in this study that shows that "partial" EMT is occurring in vivo. The rt-PCR studies presented in Fig 8 are not sufficient. Even if one accepts their incomplete analysis of transcriptomic studies using RT-PCR rather than a complete transcriptomic assessment, the study was done on bulk RNA from the entire kidney. The source material includes all cell types, not just epithelial cells, so there is no way to be sure that EMT is occurring. As noted elsewhere, they found no histologic evidence for injury and had no immunostaining results demonstrating "partial EMT" of damaged renal epithelial cells.

All of the evidence described is indirect, and the responses, while plausible, are generally excuses for lack of truly unequivocally positive results. The authors acknowledge the potential confounders of lower BP response in the Lox1-KO, unexpected weight loss in response to high fat diet, the lack of meaningful histologic evidence of injury, and they also acknowledge the absence of increased Gq signaling in the kidney, which is central to their model, but defend the entire model based on some minor changes in urinary 8-OHdG and albumin levels and a curated set of transcriptional changes. Their data could support their model—loss of Lox1 seems to reduce the levels somewhat, but the data are preliminary.

There remain serious reservations about the immunostaining results, with explanations and new data not reassuring. The authors report that they are unable to co-stain for Lox1 and AT1R because both were generated in rabbit, but this reviewer didn't ask for co-staining of the two markers. Rather, it was co-staining showing that Lox1 and AT1R in fact stain in a specific manner to the same nephron segments. The authors have added a supplementary figure showing co-staining for LOX1/AT1R with megalin, a marker for proximal tubules. However, several aspects of this are problematic:

- i. The pattern in the new Supp Fig 10 does not look like that in Fig 9. In the latter, staining is virtually everywhere, all nephron segments, and predominantly basolateral. In Supp Fig 10, they note that the pattern is primarily in the microvilli of the proximal tubule, where megalin is present. The new studies also seem to be a bit more specific, ie there are some tubules that appear to not stain with the markers.
- ii. It is difficult to be certain that the megalin staining isn't simply "bleed-through" of the signal from the other antibody. The paper doesn't describe the secondary antibody used for megalin to be sure that the emission spectra completely non-overlapping and it isn't clear that the microscope that was used offers necessary precision.
- iii. Their explanation for the pattern of AT1R staining is unconvincing. AT1R immunolocalization is known to be challenging, prompting Schrankl et al to do a definitive study using RNAscope to localize its expression in mice, rats and humans (Am J Physiol Renal Physiol 320: F644-F653, 2021). It argues against the pattern seen in Figure 9 (diffuse tubular expression), though it does suggest it is present in proximal tubules in mice. But perhaps more problematic for their model is that AT1R is not expressed in human tubules (or at least the RNA is undetectable).

Why isn't there more colocalization apparent for the AT1R and LOX1 if they form a co-receptor complex? They say that the complexes may be very dynamic, yet their movie in Suppl Fig 1 does not really support that. Not only are there few overlapping puncta in the static image, there is very little change over the duration of the movie. We don't see complexes form and then disappear and we see few new complexes form.

The explanation for why the number of replicates is variable is not reassuring. The authors note that it was because of the higher variability of the results, necessitating a higher "N" to achieve significance, but this has the appearance of P-chasing.

<https://doi.org/10.7554/eLife.98766.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

This study demonstrates a key role of oxLDL in enhancing Ang II-induced Gq signaling by promoting the AT1/LOX1 receptor complex formation. Importantly, Gq-mediated calcium influx was only observed in LOX1 and AT1 both expressing cells, and AT1-LOX1 interaction aggravated renal damage and dysfunction under the condition of a high-fat diet with Ang II infusion, so this study indicated a new therapeutic potential of AT1-LOX1 receptor complex in CKD patients with dyslipidemia and hypertension.

Strengths:

This study is very exciting and the work is also very detailed, especially regarding the mechanism of LOX1-AT1 receptor interaction and its impact on oxidative stress, fibrosis, and inflammation.

Weaknesses:

The direct evidence for the interaction between AT1 and LOX1 receptors in cell membrane localization is relatively weak. Here I raise some questions that may further improve the study.

Major points:

(1) The authors hypothesized that in the interaction of AT1/LOX1 receptor complex in response to ox-LDL and AngII, there should be strong evidence of fluorescence detection of colocalization for these two membrane receptors, both in vivo and in vitro. Although the video evidence for AT1 internalization upon complex activation is shown in Figure S1, the more important evidence should be membrane interaction and enhanced signal of intracellular calcium influx.

Thank you for your valuable feedback. We agree that demonstrating the colocalization and interaction of AT1 and LOX-1 receptors at the membrane is critical to supporting our hypothesis.

In response, we have previously provided visual evidence of membrane co-localization of the AT1/LOX-1 receptor complex using an in situ PLA assay with anti-FLAG and antiV5 antibodies in CHO cells expressing FLAG-tagged AT1 and V5-tagged LOX-1 (Yamamoto et al., FASEB J 2015). This was further supported by immunoprecipitation of membrane proteins in CHO cells co-expressing LOX-1 and AT1, which confirmed the presence of the receptor complex. In the current study, we offer additional evidence of enhanced intracellular calcium influx following simultaneous stimulation with oxLDL and Ang II, confirming the functional activation of the AT1/LOX-1 receptor complex (Fig. 1g-j and Fig. 3e-h). Together, these findings provide substantial support for the colocalization of AT1 and LOX-1 and their influence on downstream signaling in our in vitro experiments.

However, we acknowledge the limitation of direct evidence for membrane co-localization of LOX-1 and AT1 in vivo. This constraint is attributed to the fact that both available anti-AT1 and anti-LOX-1 antibodies are derived from rabbits, making coimmunofluorescence or PLA challenging in our study. To address this, we employed coimmunofluorescent staining with megalin, a well-established marker for proximal renal tubules, as shown in Fig. S10. We

found that both AT1 and LOX-1 co-localized with megalin, particularly at the brush borders, indicating their presence in the same renal compartments relevant to AT1/LOX-1 signaling.

We have revised the manuscript to highlight the functional evidence from calcium influx assays, supported by prior PLA results, demonstrating the interaction between LOX-1 and AT1. Additionally, we included a figure showing the co-localization of AT1 and LOX-1 with megalin in proximal renal tubules to reinforce these findings. Lastly, we have emphasized in the discussion the limitation regarding the lack of direct in vivo evidence for membrane co-localization of LOX-1 and AT1.

(2) Co-IP experiment should be provided to prove the AT1/LOX1 receptor interaction in response to ox-LDL and AngII in AT1 and LOX1 both expressing cells but not in AT1 only expressing cells.

We thank the reviewer for the insightful suggestion to validate the AT1/LOX1 receptor interaction under various stimulation conditions. In our previous study (Yamamoto et al., FASEB J 2015), we demonstrated the interaction between AT1 and LOX1 receptors through Co-IP and in situ PLA assays in cells overexpressing both receptors, without stimulation. These experiments provided solid evidence of the receptor interaction under static conditions at the cell membrane.

However, as noted in the previous work, we did not perform Co-IP experiments under AngII or oxLDL stimulation. The primary reason for this is that both AngII and oxLDL trigger internalization of the AT1 and/or LOX1 receptors, which may complicate the detection of receptor interaction at the membrane via Co-IP. This is supported by our realtime imaging, which showed a reduction in AT1 and/or LOX1 puncta following stimulation, indicating internalization of the receptors (Fig. 2a).

While we acknowledge the reviewer's interest in investigating the interaction under AngII stimulation, we believe that the current data—especially from the PLA and Co-IP assays under static conditions—strongly support the interaction of AT1 and LOX1 receptors at the membrane.

(3) The authors mentioned that the Gq signaling-mediated calcium influx may change gene expression and cellular characteristics, including EMT and cell proliferation. They also provided evidence that oxidative stress, fibrosis, and inflammation were all enhanced after activating both receptors and inhibiting Gq was effective in reversing these changes. However, single stimulation with ox-LDL or AngII also has strong effects on ROS production, inflammation, and cell EMT, which has been extensively proved by previous studies. So, how to distinguish the biased effect of LOX1 or AT1r alone or the enhanced effect of receptor conformational changes mediated by their receptor interaction? Is there any better evidence to elucidate this point?

Thank you for raising this important point regarding the distinction between the individual effects of LOX-1 or AT1R activation and the enhanced effects mediated by their interaction. In our study, the concentration of oxLDL used (2–10 µg/ml) was significantly lower than concentrations typically employed in other studies (which often exceed 20 µg/ml). As a result, oxLDL alone produced minimal effects, aside from a reduction in cell proliferation observed in the BrdU assay. This suggests that oxLDL, at the concentrations used in our experiments, does not elicit a strong cellular response on its own.

The key to distinguishing the effect of the LOX-1/AT1 interaction lies in the amplification of Gq signaling, a pathway specifically activated by AngII. The distinction between the individual effects of LOX-1 or AT1R and the enhanced effects due to their interaction is centered on the increased activation of Gq signaling. In our experiments, co-treatment with

oxLDL and AngII led to a significant increase in IP1 levels and calcium influx— both critical indicators of Gq signaling activation. While AngII alone also raised IP1 levels, the combined treatment with oxLDL further amplified the Gq signaling response, as reflected in the enhanced calcium influx. Importantly, oxLDL alone did not alter IP1 levels, even at high concentrations (100 µg/ml) (Takahashi et al., iScience 2021).

This enhancement of Gq signaling provides strong evidence of the synergistic interaction between LOX-1 and AT1, which surpasses the individual effects of either receptor alone. The LOX-1/AT1 interaction is thus crucial for the observed amplification of AngII-specific signaling pathways. The combination of increased IP1 levels and calcium influx serves as compelling evidence of this interaction, clearly differentiating the effects of individual receptor activation from the enhanced response driven by receptor conformational changes and interaction.

Thank you again for your insightful comment, which has helped us to better articulate the significance of receptor interaction in this study.

(4) How does the interaction between AT1 and LOX1 affect the RAS system and blood pressure? What about the serum levels of rennin, angiotensin, and aldosterone in ND-fed or HFD-fed mice?

Thank you for your insightful question regarding the effects of AT1 and LOX-1 interaction on the renin-angiotensin system (RAS) and blood pressure, as well as the plasma levels of renin, angiotensin, and aldosterone in normal diet (ND)-fed and high-fat diet (HFD)-fed mice.

OxLDL binds to LOX-1, amplifying AT1 receptor activation and Gq signaling, which enhances the effects of Ang II. This interaction between AT1 and LOX-1 can lead to increased vasoconstriction, oxidative stress, and inflammation, which contribute to elevated blood pressure. This pathway may play a crucial role in modulating the RAS, particularly under conditions of elevated oxLDL, such as those induced by a HFD. Regarding the components of the RAS, we focused on plasma aldosterone levels, as this is a direct consequence of Ang II signaling. As shown in Fig. S7, when mice were treated with a pressor dose of Ang II infusion and subjected to a HFD to elevate oxLDL levels, we did not observe a significant increase in plasma aldosterone levels (102.8 ± 11.6 pg/mL vs. 141.8 ± 15.0 pg/mL, $P = 0.081$).

In terms of blood pressure, Fig. 7b shows that no significant changes were observed under these treatment conditions, despite the AT1/LOX-1 interaction. These findings suggest that while oxLDL, via the AT1/LOX-1 interaction, can enhance Ang II signaling, its effect on blood pressure was not apparent in our study. This may be due to several factors, including heterogeneous cellular responses to the combined treatment across different cell types, as shown by the lack of reaction in vascular endothelial cells, vascular smooth muscle cells, and macrophages (Fig. S2). This may also be attributed to the high concentration of angiotensin II used in this study, which could have saturated aldosterone production under our experimental conditions. We have revised the manuscript to reflect these points.

Thank you again for your thoughtful comment, which has allowed us to expand and refine the discussion on this important aspect of our study.

Reviewer #2 (Public Review):

(1) Individuals with chronic kidney disease often have dyslipidemia, with the latter both a risk factor for atherosclerotic heart disease and a contributor to progressive kidney disease. Prior studies suggest that oxidized LDL (oxLDL) may cause renal injury through the activation of the LOX1 receptor. The authors had previously reported that LOX1 and AT1 interact to form a complex at the cell surface. In this study, the authors hypothesize that oxLDL, in the setting of angiotensin II, is responsible for driving renal injury by

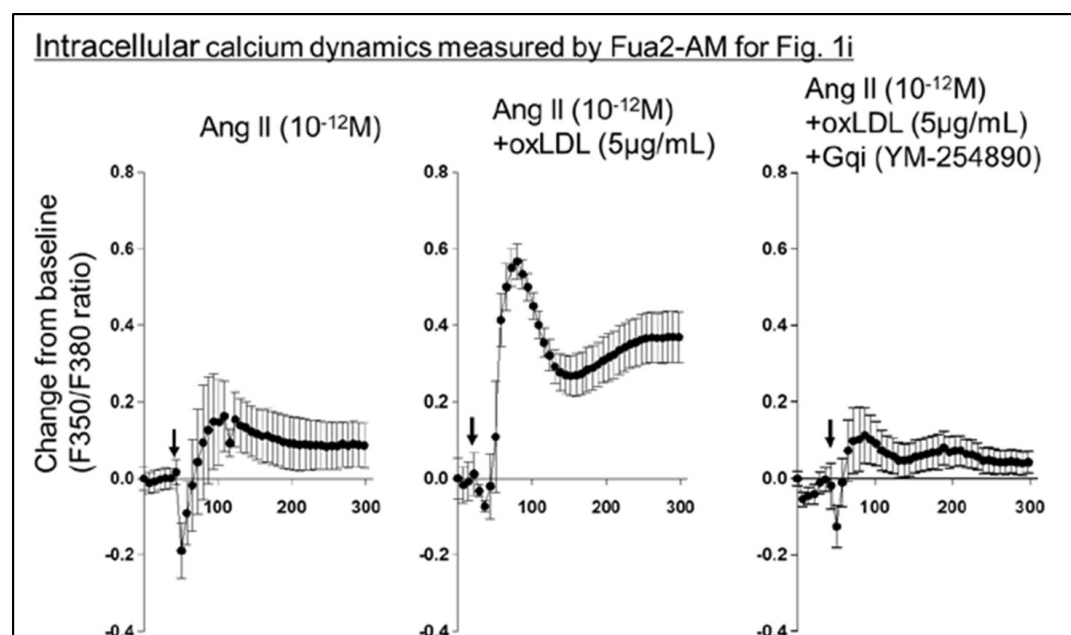
inducing a more pronounced conformational change of the AT1 receptor which results in enhanced Gq signaling.

They go about testing the hypothesis in a set of three studies. In the first set, they engineered CHO cell lines to express AT1R alone, LOX1 in combination with AT1R, or LOX1 with an inactive form of AT1R and indirectly evaluated Gq activity using IP1 and calcium activity as read-outs. They assessed activity after treatment with AngII, oxLDL, or both in combination and found that treatment with both agents resulted in the greatest level of activity, which could be effectively blocked by a Gq inhibitor but not a Gi inhibitor nor a downstream Rho kinase inhibitor targeting G12/13 signaling. These results support their hypothesis, though variability in the level of activation was dramatically inconsistent from experiment to experiment, differing by as much as 20-fold. In contrast, within the experiment, differences between the AngII and AngII/oxLDL treatments, while nominally significant and consistent with their hypothesis, generally were only 10-20%. Another example of unexplained variability can be found in Figures 1g-1j. AngII, at a concentration of 10⁻¹², has no effect on calcium flux in one set of studies (Figure 1g, h) yet has induced calcium activity to a level as great as AngII + oxLDL in another (Figure 1i). The inconsistency of results lessens confidence in the significance of these findings. In other studies with the LOX1-CHO line, they tested for conformational change by transducing AT1 biosensors previously shown to respond to AngII and found that one of them in fact showed enhanced BRET in the setting of oxLDL and AngII compared to AngII alone, which was blocked by an antibody to AT1R. The result is supportive of their conclusions. Limiting enthusiasm for these results is the fact that there isn't a good explanation as to why only 1 sensor showed a difference, and the study should have included a non-specific antibody to control for non-specific effects.

We sincerely appreciate the reviewer's thorough and insightful feedback, especially regarding the variability observed in our experimental results. As the reviewer pointed out, the differences in activation levels between the calcium influx assay and the IP1 assay, particularly between AngII and AngII/oxLDL co-treatment, were indeed significant. These differences can be attributed to the inherent sensitivity of these assays, which are used to indirectly evaluate Gq activity. Despite the variability, we believe that the reliability of our results is supported by the consistent directional trends across both assays, which align with our hypothesis.

Regarding the inconsistencies in intracellular calcium dynamics observed in Fig. 1i, we have performed additional analysis of calcium kinetics during ligand stimulation, similar to the analysis in Fig. 1g. As shown in Author response image 1, the background signal in the experiment related to Fig. 1i was relatively higher than in Fig. 1g and 1h. This elevated background, which may have been influenced by variations between cells and experimental days, resulted in a higher percent change from baseline in samples treated with AngII alone. However, the combined effect of AngII with oxLDL was still apparent. This clarification further supports the consistency of our findings.

Author response image 1.



In reference to the BRET sensor experiments, we acknowledge the reviewer's concern regarding the variability in sensor responses. As outlined in Devost et al. (J Biol Chem. 2017), the sensitivity of AT1 intramolecular FAsH-BRET biosensors in detecting conformational changes induced by AngII is highly dependent on the insertion site of the FAsH sequence. In our experiments, co-treatment with oxLDL and AngII enhanced AT1 conformational changes, but this effect was only detectable with the CHO-LOX-1-AT1-3p3 sensor (with FAsH inserted in the third intracellular loop), and not with the CHO-LOX-1-AT1-C-tail P1 sensor (with FAsH inserted at the C-terminal tail). This differential sensitivity likely explains why only one sensor showed a significant response, highlighting the critical role of FAsH insertion site selection in these assays. We hope these clarifications address the reviewer's concerns and improve confidence in the significance of our findings.

(2) The authors then repeated similar studies using publicly available rat kidney epithelial and fibroblast cell lines that have an endogenous expression of AT1R and LOX1. In these studies, oxLDL in combination with AngII also enhanced Gq signaling, while knocking down either AT1R or LOX1, and treatment with inhibitors of Gq and AT1R blocked the effects. Like the prior set of studies, however, the effects are very modest and there was significant inter-experimental variability, reducing confidence in the significance of the findings. The authors then tested for evidence that the enhanced Gq signaling could result in renal injury by comparing qPCR results for target genes. While the results show some changes, their significance is difficult to assess. A more global assessment of gene expression patterns would have been more appropriate. In parallel with the transcriptional studies, they tested for evidence of epithelial-mesenchymal transition (EMT) using a single protein marker (alpha-smooth muscle actin) and found that its expression increased significantly in cells treated with oxLDL and AngII, which was blocked by inhibition of Gq inhibition and AT1R. While the data are sound, their significance is also unclear since EMT is a highly controversial cell culture phenomenon. Compelling in vivo studies have shown that most if not all fibroblasts in the kidney are derived from interstitial cells and not a product of EMT. In the last set of studies using these cell lines, the authors examined the effects of AngII and oxLDL on cell proliferation as assayed using BrdU. These results are puzzling---while the two agents together

enhanced proliferation which was effectively blocked by an inhibitor to either AT1R or Gq, silencing of LOX1 had no effect.

Thank you for your thorough review and comments. We acknowledge your concerns regarding the modest effects observed and the variability in experimental outcomes. We would like to address your points systematically.

(1) Gq signaling and experimental variability:

Regarding the question of Gq signaling in Fig. 3, as previously mentioned, the observed differences in the IP1 assay are likely due to the sensitivity of the assay and the technical issues associated with detecting calcium influx and IP1 levels. While the overall differences between treatments may appear modest, the most critical comparison—between AngII alone and AngII combined with oxLDL—consistently showed significant differences, which aligns with the calcium influx results shown in Fig. 1. Notably, we found that the EC50 for IP1 production decreased by 80% in response to co-treatment with oxLDL and AngII, compared to AngII treatment alone. These findings demonstrate the robustness of Gq signaling enhancement with co-treatment, even if the absolute differences in the IP1 assay appear small.

(2) Gene expression in Fig. 4:

Regarding the gene expression analysis in Fig. 4, we used relatively low concentrations of oxLDL (5 µg/ml) compared to the higher concentrations typically employed in other studies (mostly exceeding 20 µg/ml). This may explain the lack of robust responses in some conditions. However, in combination with AngII, the co-treatment significantly upregulated several genes, particularly pro-inflammatory markers such as IL-6, TNFα, IL1β, and MCP-1 in NRK49F cells. These results suggest that the co-treatment induces a complex response, potentially activating multiple downstream signaling pathways beyond just Gq signaling, which may obscure more straightforward effects.

While we agree that a more global assessment of gene expression would provide further insights, due to cost constraints, we focused on key representative genes that are highly relevant to inflammation and fibrosis in this study.

(3) EMT in renal fibrosis:

We appreciate the reviewer's insightful comments regarding the role of EMT in renal fibrosis. Regarding full EMT, in which epithelial cells completely transition into mesenchymal cells, previous studies using the unilateral ureteral obstruction (UUO) model suggest that full EMT may not play a significant role (J Clin Invest. 2011 Feb;121(2):468-74). The role of full EMT remains controversial in the context of renal fibrosis, with most kidney fibroblasts thought to originate from interstitial cells rather than through full EMT.

Recent studies, however, suggest that partial epithelial-mesenchymal transition (pEMT) could be involved in CKD, especially in association with inflammation, oxidative stress, and elevated TGF-β levels—conditions also present in our model involving Ang II infusion combined with an HFD. pEMT refers to a state in which epithelial cells acquire mesenchymal traits, such as increased α-SMA expression and secretion of pro-fibrotic cytokines, while remaining attached to the basement membrane without fully transitioning into fibroblasts (Front Physiol. 2020 Sep 15;11:569322). This phenomenon has been observed in kidney fibrosis models, including UUO, which shares inflammatory and oxidative stress conditions with our Ang II and HFD treatment model. The observed increase in α-SMA in our model may thus indicate a pEMT-like state, indirectly contributing to fibrosis through the secretion of growth factors and cytokines.

We are mindful of the importance of not overstating EMT's role. Accordingly, we interpret increased α -SMA expression as a potential marker of the pEMT process rather than definitive evidence of its presence or direct role in fibroblast formation. Furthermore, we acknowledge limitations in providing direct in vivo evidence for pEMT and recognize that further mechanistic studies are needed to elucidate its specific role in renal fibrosis, despite inherent challenges.

In response to the reviewer's concern, we have revised the manuscript to clarify that our data support the possibility of pEMT contributing to fibrosis in this model, without overstating its impact. We also acknowledge the challenges in translating in vitro pEMT findings to in vivo models, where detecting the subtle effects of pEMT is inherently challenging.

(4) BrdU assay and fibroblast proliferation (Fig. 6b):

In Fig. 6b, the BrdU assay shows that fibroblast proliferation was significantly enhanced by the co-treatment with AngII and oxLDL, and this effect was abolished by LOX-1 knockdown, similar to the results observed with AT1 knockdown. These findings strongly suggest a combinatorial effect of AT1/LOX-1 interaction in promoting fibroblast proliferation, supporting the idea that the co-treatment operates through a coordinated mechanism involving both receptors. Notably, LOX-1 silencing did not affect the proliferation induced by AngII alone, as this response is independent of LOX-1.

We will incorporate these points into the Discussion section of the manuscript, specifically regarding the differences in sensitivity between the Ca influx and IP1 assays, as well as the emerging role of partial EMT in renal fibrosis. This will provide a clearer context for the interpretation of our findings and further strengthen the discussion on the significance of these phenomena.

Thank you again for your valuable feedback, which has helped us improve the clarity and depth of our manuscript.

(3) The final set of studies looked to test the hypothesis in mice by treating WT and Lox1KO mice with different doses of AngII and either a normal or high-fat diet (to induce oxLDL formation). The authors found that the combination of high dose AngII and a highfat diet (HFD) increased markers of renal injury (urinary 8-ohdg and urine albumin) in normal mice compared to mice treated with just AngII or HFD alone, which was blunted in Lox1-KO mice). These results are consistent with their hypothesis. However, there are other aspects of these studies that are either inconsistent or complicating factors that limit the strength of the conclusions. For example, Lox1- KO had no effect on renal injury marker expression in mice treated with low-dose AngII and HFD. It also should be noted that Lox1-KO mice had a lower BP response to AngII, which could have reduced renal injury independent of any effects mediated by the AT1R/LOX1 interaction. Another confounding factor was the significant effect the HFD diet had on body weight. While the groups did not differ based on AngII treatment status, the HFD consistently was associated with lower total body weight, which is unexplained. Next, the authors sought to find more direct evidence of renal injury using qPCR of candidate genes and renal histology. The transcriptional results are difficult to interpret; moreover, there were no significant histologic differences between groups. They conclude the study by showing the pattern of expression of LOX1 and AT1R in the kidney by immunofluorescence and conclude that the proteins overlap in renal tubules and are absent from the glomerulus. Unfortunately, they did not co-stain with any other markers to identify the specific cell types. However, these results are inconsistent with other studies that show AT1R is highly expressed in mesangial cells, renal interstitial cells, near the vascular pole, JG cells, and proximal tubules but generally absent from most other renal tubule segments.

Thank you for your valuable comments and for raising these important points. We appreciate the opportunity to clarify several aspects of our study and address the limitations and inconsistencies you have pointed out.

(1) Renal injury markers (urinary albumin and 8-OHdG) and the effect of LOX-1 loss of function:

Our results showed that the combination of high-dose AngII and HFD led to a significant increase in renal injury markers, such as urinary albumin and 8-OHdG, in WT mice. In LOX-1 KO mice, this increase was significantly blunted, supporting a protective role of LOX-1 loss-of-function. However, as you noted, at low-dose AngII, there was no significant difference in urinary 8-OHdG between ND-fed and HFD-fed mice. Despite this, we observed a significant increase in urinary albumin in HFD-fed WT mice compared to ND-fed mice under low-dose AngII, and this difference was abolished in LOX-1 KO mice. Moreover, gene expression analysis showed that oxidative stress markers such as p67phox and p91phox (Fig. 8b), as well as p40phox, p47phox (Fig. S8), and inflammatory markers like IL1 β (Fig. 8b), were significantly elevated in HFD-fed WT mice even with low-dose AngII, while these increases were absent in LOX-1 KO mice. These results suggest that the LOX-1/AT1 interaction contributes to renal injury under both low- and high-dose AngII conditions.

We acknowledge that the treatment duration may have influenced our results, as urine and renal tissue samples were only examined at a single time point (1.5 months after treatment initiation). The impact of AT1/LOX-1 interaction may evolve over time, and different treatment durations might yield varying outcomes. This is a limitation of our study, which we have addressed in the revised manuscript.

(2) Blood pressure and its effect on renal injury:

As shown in Fig. 7b and Fig S6f, LOX-1 KO mice exhibited a lower blood pressure response to high-dose AngII compared to WT mice, which could indeed have contributed to the reduced renal injury in the LOX-1 KO group, independent of the AT1/LOX-1 interaction. However, it is important to note that the differences in renal injury markers between AngII alone and AngII + HFD were largely abolished in LOX-1 KO mice, suggesting the *in vivo* relevance of the LOX-1/AT1 interaction observed *in vitro*. Additionally, as shown in Fig. 7d (urinary albumin), Fig. 8b (p67phox, p91phox), and Fig. S8b (p40phox, p47phox), even under suppressor doses of AngII, where no significant blood pressure differences were observed, HFD-fed WT mice exhibited exacerbated renal injury compared to ND-fed mice. These effects were ameliorated in LOX-1 KO mice, indicating that the protective effects in LOX-1 KO mice are at least partly independent of blood pressure changes and that the AT1/LOX-1 interaction plays a significant role in modulating renal injury under co-treatment with AngII and HFD.

(3) HFD and body weight changes:

We agree with your observation regarding the effect of HFD on body weight, which was consistently lower in HFD-fed groups, despite no differences in AngII treatment status. This is an atypical presentation compared to previous studies mostly showing increased body weight by feeding of HFD. The HFD used in this study was intended to elevate oxLDL levels, as previously reported (Atherosclerosis 200:303–309 (2008)). As shown in Fig. S6d and S6e, this can be attributed to reduced food intake in HFD-fed mice. Although modest, this weight reduction may influence renal function. This point is added in the limitation.

(4) Histological findings and qPCR results:

As discussed in the manuscript, despite significant changes in urinary markers and gene expression, we did not observe histological evidence of fibrosis or mesangial expansion, even under co-treatment with AngII and HFD. This may be due to the relatively short treatment

period of 4 weeks, and a longer duration might be necessary to detect such changes. Additionally, we acknowledge that we did not detect increased Gq signaling in kidney tissue, which is another limitation of the study. Nevertheless, the gene expression data on oxidative stress, fibrosis, inflammation, and renal injury markers (e.g., p67phox, IL1 β) are consistent with our hypothesis that the AT1/LOX-1 interaction exacerbates renal injury under AngII and HFD conditions.

(5) Immunostaining for AT1 and LOX-1:

Due to the use of rabbit-derived antibodies for both AT1 and LOX-1, it was technically not feasible to perform co-immunostaining for both receptors simultaneously. Instead, we performed co-immunofluorescent staining using megalin, a well-established marker of proximal renal tubules, to help localize these receptors. As shown in Fig. S10, both AT1 and LOX-1 were co-localized with megalin, particularly at the brush borders of proximal tubules. This pattern suggests the presence of these receptors in renal compartments relevant to AT1/LOX-1 signaling. While we did not perform additional co-staining with other markers to identify specific cell types, the strong localization with megalin provides robust evidence of their expression in proximal renal tubules, which is consistent with the literature on AT1R in this nephron segment. We acknowledge that previous studies have identified AT1R expression in mesangial cells, renal interstitial cells, the vascular pole, juxtaglomerular (JG) cells, and proximal tubules. In our immunofluorescence experiments, we did not detect significant AT1R expression in the glomerulus or mesangium. This finding aligns with other reports showing strong expression of AT1R in proximal tubules (Am J Physiol Renal Physiol. 2021 Apr 1;320(4)), although it does not exclude the possibility of AT1 expression in other compartments, given the sensitivity limitations of the immunofluorescence. Our focus on proximal tubules allowed us to observe clear AT1/LOX-1 co-localization in this region, particularly in the context of oxLDL and AngII signaling. Given that the AT1/LOX-1 interaction is crucial in kidney disease pathogenesis, this co-localization in proximal tubules highlights a key site of action for these receptors in the renal system.

In summary, while our study focused on the co-localization of AT1 and LOX-1 in proximal tubules, we agree that further exploration of AT1R expression in other renal cell types would provide a more comprehensive understanding of its role across different kidney compartments. We have addressed this in the revised discussion.

Reviewer #1 (Recommendations For The Authors):

Minor points:

(1) In this study, AT1/LOX1 receptor complex was mainly observed in some renal cells, how about other types of cells that also highly express LOX1 and AT1r? Such as cardiomyocytes? Vascular endothelial cells?

Thank you for your insightful comment. In our study, we demonstrated that enhanced Gq signaling through co-treatment with AngII and oxLDL was not observed in other cell types, including vascular endothelial cells, smooth muscle cells, and macrophages, as indicated by the lack of an IP1 increase in response to the co-treatment (Fig. S2). The factors contributing to this heterogeneous response remain unclear, and further investigation is needed to explore this observation more thoroughly.

(2) Has the author detected such an effect on the AT2 receptor?

We greatly appreciate the reviewer's insightful inquiry regarding the potential interaction between the AT2 receptor and LOX-1. In our previous work (Yamamoto et al., FASEB J 2015), we conducted an immunoprecipitation (IP) assay to investigate the interaction between LOX-1 and AT2 on cell membranes. The results of this assay demonstrated that, unlike AT1, LOX-1

exhibits minimal binding to the AT2 receptor under the experimental conditions tested. Specifically, our IP studies showed that while LOX-1 readily coimmunoprecipitated with AT1, indicating a strong interaction, this was not the case with AT2, where the binding was negligible. These findings suggest that the interaction between LOX-1 and AT1 is receptor-specific and that LOX-1 does not significantly associate with AT2 to influence signaling pathways.

(3) Which kind of ARBs are more effective for the inhibition of this AT1/LOX1 receptor conformational change?

Thank you for your insightful question regarding the effectiveness of ARBs in inhibiting the AT1/LOX-1 receptor conformational change. Based on our current understanding, any ARB should similarly block the downstream signaling resulting from the interaction between AT1 and LOX-1. This is because all ARBs function by inhibiting the binding of Ang II to AT1, thereby preventing receptor activation and the conformational changes that facilitate its interaction with LOX-1. Additionally, our previous study (FASEB J. 2015) demonstrated that even in the absence of Ang II, the activation of AT1 via the binding of oxLDL to LOX-1 was similarly blocked by ARBs, including olmesartan, telmisartan, valsartan, and losartan.

When oxLDL and Ang II are co-treated, the Gq signaling pathway is significantly amplified due to the interaction between LOX-1 and AT1. In this setting, all ARBs act by competitively inhibiting Ang II binding to AT1, effectively reducing Gq signaling.

However, a subtle but important difference arises when considering the inverse agonist activity of certain ARBs. Olmesartan, telmisartan, and valsartan are thought to act not only as competitive inhibitors of Ang II but also as inverse agonists, meaning they reduce the baseline activity of the AT1 receptor by preventing the conformational changes in the absence of Ang II. This inverse agonist property is particularly relevant in pathological conditions where AT1 receptor activation can occur independently of Ang II binding, such as in the presence of oxLDL. In these cases, ARBs with inverse agonist activity may offer an additional therapeutic advantage by reducing receptor activation beyond what is achieved by simple antagonism.

Thus, while the general efficacy of ARBs in blocking the AT1/LOX-1 interaction could be under similar conditions of oxLDL and Ang II co-treatment, ARBs with inverse agonist properties may provide additional benefit by further reducing AT1 activity.

We have revised the manuscript to clarify these points and to highlight the role of inverse agonist activity in ARB efficacy under these conditions.

Thank you again for your valuable comment, which has allowed us to refine our discussion on the relative efficacy of ARBs in inhibiting AT1/LOX-1 receptor interaction.

Reviewer #2 (Recommendations For The Authors):

My comments were pretty thorough in the public review. The only other comments I would add are the following:

(1) Why are there so few overlapping LOX1 and ATR puncta in Supplementary Figure 1 if the receptors co-localize? The figure would suggest a very small proportion of the receptors actually are co-localized.

Thank you for your insightful comment regarding the apparent scarcity of overlapping LOX-1 and AT1R puncta in Fig. S1. We agree that at first glance, the low number of colocalized puncta may raise questions about the extent of interaction between these receptors.

However, based on our previous findings reported in FASEB J 2015, we believe this phenomenon can be explained by the dynamic nature of the LOX-1 and AT1 interaction.

As we reported in FASEB J 2015, the interaction between LOX-1 and AT1 is sensitive to buffer conditions. Specifically, in non-reducing conditions, LOX-1 and AT1 form complexes, whereas in reducing buffer, this interaction is not observed. This suggests that the interaction between these receptors is not stabilized by strong covalent (disulfide) bonds but is instead transient, likely involving non-covalent interactions. Thus, LOX-1 and AT1 may form and dissociate repeatedly, contributing to a dynamic receptor complex rather than a permanent colocalization. This transient interaction could explain the relatively low number of overlapping puncta observed at a given time point in the liveimaging analysis.

Moreover, as you pointed out, it is likely that only a small fraction of LOX-1 and AT1 are physically co-localized at any one moment. However, when these receptors do interact, co-treatment with oxLDL and Ang II has been shown to significantly enhance Gq signaling. This suggests that the functional consequence of the LOX-1/AT1 interaction, particularly in response to stimuli such as oxLDL and Ang II, is more critical than the frequency of receptor colocalization at any one time.

We have revised the manuscript to include this explanation and to clarify the dynamic nature of the LOX-1/AT1 interaction. This revision also highlights the importance of considering not just the number of colocalized receptors but also the functional outcomes of their interaction, such as enhanced Gq signaling in response to co-treatment.

Thank you again for your careful observation, which has allowed us to better communicate the complexity of the receptor dynamics in our study.

| (2) *Tubulin is misspelled in Figure 5 ("tublin").*

Thank you for pointing out the typographical error in Fig. 5. We have corrected the spelling of "tubulin" in the revised figure. We appreciate your attention to detail, and we apologize for the oversight.

| (3) *Why does the number of replicates differ for some experimental sets (i.e. Figure 1h vs other panels in Figure 1, Figure 2d vs other panels in Figure 2, Figure 7: Lox-1KO treated with High dose AngII and HFD)? There aren't obvious reasons why the number of replicates should differ so much within a set of studies.*

We are grateful to the reviewer for highlighting the discrepancies in the number of replicates across different figures in our manuscript. We would like to provide detailed explanations for each case.

(1) Fig. 1h vs Other Panels in Fig. 1:

The calcium influx assay (Fig. 1h) required a higher number of replicates due to the inherent biological variability associated with calcium signaling. To achieve statistical significance and account for variability in these measurements, we conducted additional replicates. Other panels, such as those measuring IP1 accumulation (Fig. 1a–f), displayed more consistent and reproducible results, allowing us to use fewer replicates while still maintaining statistical power.

(2) Fig. 2d vs Fig. 2b and 2c:

The difference in the number of replicates between Fig. 2d (N=8) and Fig. 2b and 2c (N=4) is due to the distinct nature of the measurements and the variability expected in each assay. In Fig. 2d, which measures the effects of a LOX-1 neutralizing antibody on BRET, additional replicates were needed to ensure the robustness of the statistical analysis due to the greater

complexity and sensitivity of the assay. The inclusion of an antibody treatment introduces more variability, necessitating a higher number of replicates (N=8) to confidently assess the effects of the neutralizing antibody. In contrast, Fig. 2b and 2c involved BRET measurements of AT1 conformational changes without antibody intervention. These assays are more reproducible and have less experimental variability, allowing for a smaller sample size (N=4) while still achieving reliable and statistically significant results. The differences in sample size across these panels were carefully considered to ensure appropriate statistical power for each specific experimental condition.

(3) Fig. 7: LOX-1 KO Mice Treated with High-dose AngII vs Saline:

We acknowledge the reviewer's concern regarding the higher number of LOX-1 KO mice treated with high-dose Ang II compared to the saline group. The number of saline-treated mice was indeed sufficient for reliable statistical analysis. However, the decision to increase the number of mice in the high-dose Ang II group was driven by the anticipated higher variability in the physiological responses under these conditions, such as blood pressure and renal injury. To ensure that we captured the full spectrum of responses and to maintain robust statistical power in the high-dose group, we opted to include more mice in this cohort.

We hope this response provides clarity on the rationale behind the varying number of replicates across different experiments. We have rigorously applied appropriate statistical methods to account for these differences, ensuring that the conclusions drawn are robust and scientifically sound. We appreciate the reviewer's understanding of the experimental constraints and variations that can arise in complex studies such as these.

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